Chapter 3
Static Electric Fields

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Contents

3.1 Introduction

Fundamental Physical Quantities:

Source; Field

Source: Charge $q(r',t)$ **, Current** $I(r',t)$

$$\begin{tikzcd} \text{Charge} \arrow[r, "\text{Movement}"'] \arrow[d] & \text{Current} \arrow[d] \\ \text{Electric Field} & \text{Magnetic field} \end{tikzcd}$$

What is the model for charge?

1. Charge and Charge Density

- **Charge is one of basic elements of substance**
- **In 1897, Joseph John Thomson found charge in experiment in Britain.**
- **Between 1907 — 1913，Robert Andrews Millikan precisely measured the charge of electron through Oil Drop experiment, the charge is:**

$$e = 1.602\ 177\ 33 \times 10^{-19} \quad \text{(Unit: C)}$$

2. Form of the existed Charge（Four）：

Volume Charge、Surface Charge、Line Charge、Point Charge

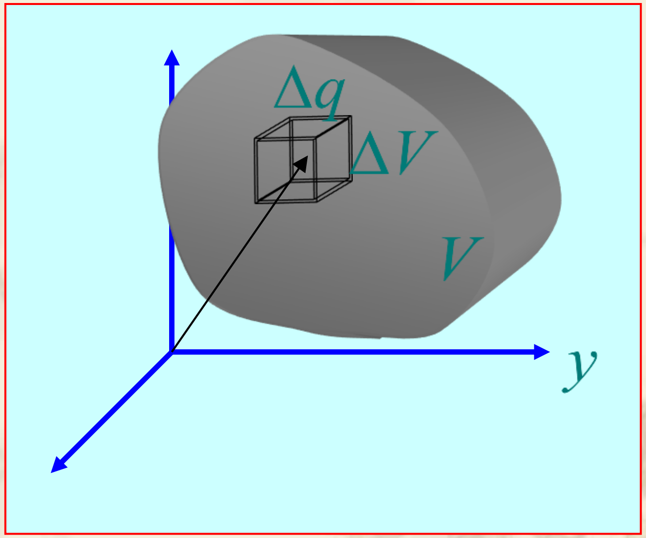
(1). Volume Charge Density

$$\rho(r)=\lim_{\Delta V\to 0}\frac{\Delta q(r)}{\Delta V}=\frac{\mathrm{d}q(r)}{\mathrm{d}V}$$

Unit: C/m³

Relation between total charge *q* and density:

$$q=\int_V \rho(r)\mathrm{d}V$$



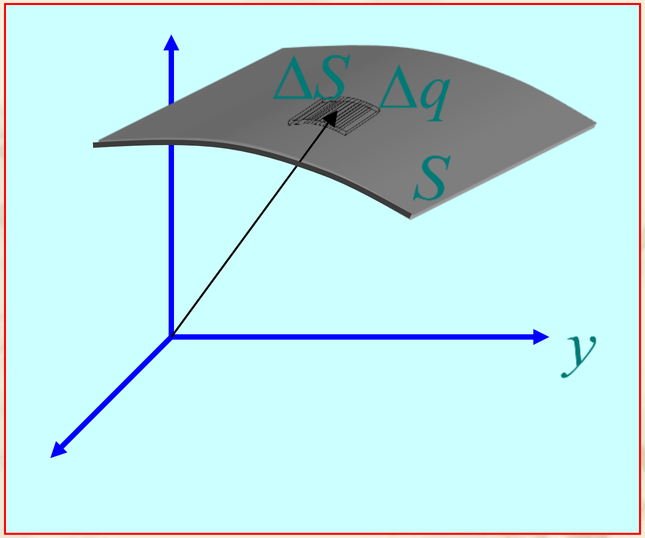
(2). Surface Charge Density

$$\rho_S(r) = \lim_{\Delta S \to 0} \frac{\Delta q(r)}{\Delta S} = \frac{\mathrm{d}q(r)}{\mathrm{d}S}$$

Unit: C/m^2

If the surface charge density on a certain surface is known, the total charge *q* on this surface is:

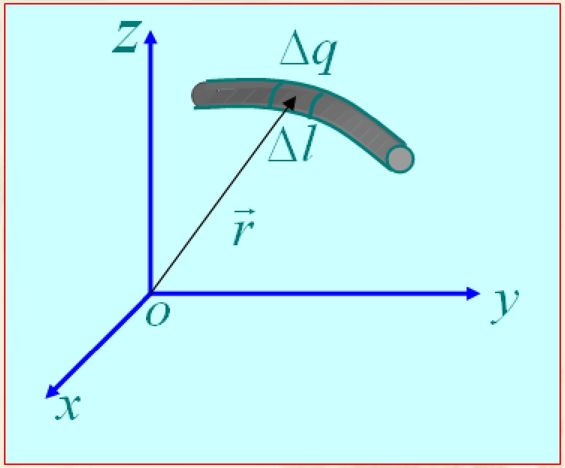
$$q = \int_S \rho_s(r)\mathrm{d}S$$



(3). Line Charge Density

$$\rho_l(r) = \lim_{\Delta l \to 0} \frac{\Delta q(r)}{\Delta l} = \frac{\mathrm{d}q(r)}{\mathrm{d}l}$$

Unit: C / m



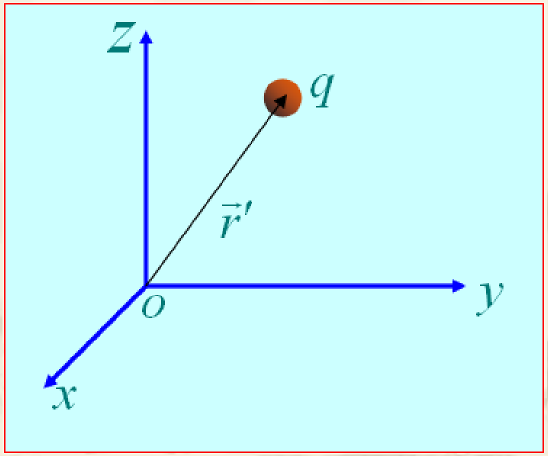
If the line charge density on a certain line is known, the total charge *q* on this line is:

$$q = \int_C \rho_l(r)\mathrm{d}l$$

(4). Point Charge

The density of point charge

$$\rho(r) = q\delta(r - r')$$

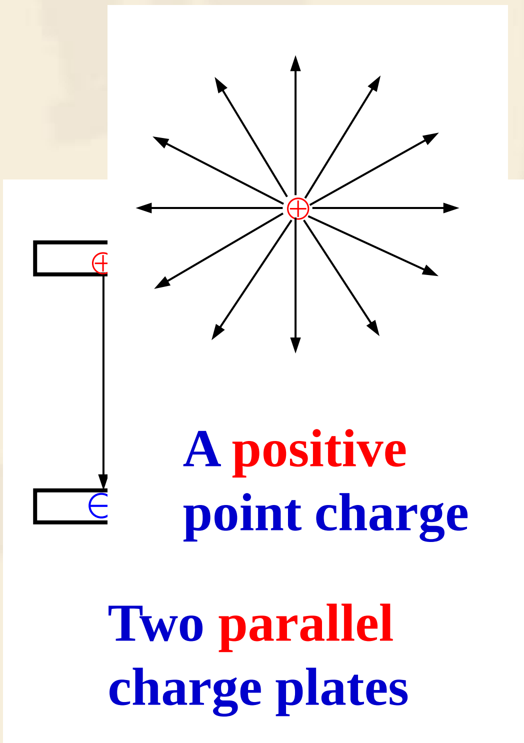


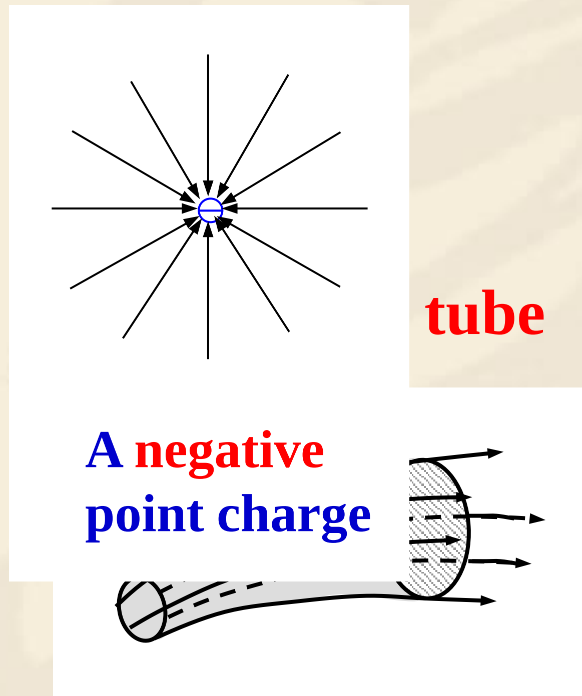
- **Unit point charge: volume→0; density →∞, volume·density=1**
- **Delta function: the density of unit point charge**

$$\int_V \delta(r - r')dV = 1$$

3.2 Fundamental Postulates of Electrostatics in Free Space

Distributions of electric field lines





3.2 Fundamental Postulates of Electrostatics in Free Space

❖ Electric field intensity and Electric Flux

• **Electric field intensity is defined as the force per unit charge**

$$\mathbf{E} = \lim_{q \to 0} \frac{\mathbf{F}}{q} \quad (v/m) \qquad\qquad F = qE \quad (N)$$

• **The flux of the electric field intensity through a surface is called electric flux, and it is denoted as , given by** Ψ**, i.e.**

$$\Psi = \int_S \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S}$$

Two Fundamental Postulates of Electrostatics in Free Space

$$\nabla \times \bar{E} = 0$$

Fields are irrotational

$$\nabla \cdot \bar{E} = \frac{\rho}{\varepsilon_0}$$

Field is not solenoidal unless $\rho = 0$

where ρ is the volume charge density of free charges (C/m3), and ε_0 is the permittivity of free space, universal constant.

- **Two postulates are concise, simple, and independent of any coordinate system; and they can be used to derive all other relations, laws and theorems in electrostatics!**
- **Their integral form can be obtained by using Stokes theorem and Divergence theorem as**

$$\nabla \times \bar{E} = 0 \Rightarrow \oint_C \bar{E} \cdot d\bar{l} = 0$$

Fields form of KVL

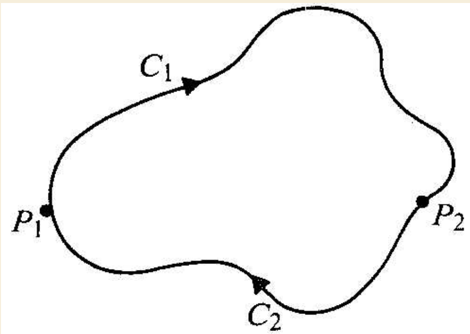
$$\nabla \cdot \bar{E} = \frac{\rho}{\varepsilon_0} \Rightarrow \oint_S \bar{E} \cdot d\bar{s} = \frac{Q}{\varepsilon_0}$$

Fields form of KCL

- **Gauss's law:** **the total outward flux of the electric field intensity over any closed surface in free space is equal to the total charge enclosed in the surface divided by** ε_0

$$\oint_S \vec{E}(\vec{r})\cdot \mathrm{d}\vec{S} = \frac{1}{\varepsilon_0} Q\Big|_{S内}$$

- **The scalar line integral of the static electric field intensity around any closed path vanishes. It is independent of the path; it depends only on the end points.**



$$\int_{C_1} \mathbf{E}\cdot d\ell + \int_{C_2} \mathbf{E}\cdot d\ell = 0$$

$$\underset{\text{Along } C_1}{\int_{P_1}^{P_2}} \mathbf{E}\cdot d\ell = - \underset{\text{Along } C_2}{\int_{P_2}^{P_1}} \mathbf{E}\cdot d\ell$$

$$\oint_C \vec{E}(\vec{r})\cdot \mathrm{d}\vec{l} = 0$$

Q: What is the difference between integral form and Differential form?

Conclusion

Divergence of electrostatic field

（ Differential form ）

$$\nabla\cdot\vec{E}(\vec{r})=\frac{\rho(\vec{r})}{\varepsilon_0}$$

Flux of electrostatic field

Gauss' law （ Integral form ）

$$\oint_S\vec{E}(\vec{r})\cdot\mathrm{d}\vec{S}=\frac{1}{\varepsilon_0}Q\Big|_{S内}$$

Curl of electrostatic field

Curl of electrostatic field

（ Differential form ）

$$\nabla\times\vec{E}(\vec{r})=0$$

Circulation of electrostatic field

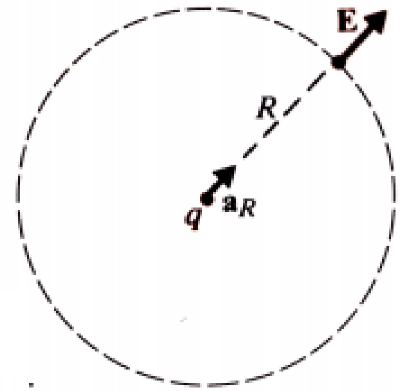
（ Integral form ）

$$\oint_C\vec{E}(\vec{r})\cdot\mathrm{d}\vec{l}=0$$

Conclusion ：

1. **Electrostatic field is irrotational and conservative, so the work of electric field forces has nothing to do with the path;**
2. **Electrostatic field is radical, starts from positive charges and ends to negative charges.**

3.3 Coulomb's Law



- A single point charge, Q, sits in origin of a boundless free space, its E field must be everywhere radial and has the same intensity at all points on the spherical surface.

$$\oint_S \bar{E}\cdot d\bar{s} = \oint_S \hat{a}_R E_R \cdot \hat{a}_R ds = \frac{Q}{\varepsilon_0} \quad \text{or} \quad E_R \oint_S ds = E_R(4\pi R^2) = \frac{Q}{\varepsilon_0}$$

Therefore,

$$\bar{E} = \hat{a}_R E_R = \hat{a}_R \frac{Q}{4\pi\varepsilon_0 R^2} \quad (V/m)$$

• When a point charge Q2 is placed in the field of another point charge Q1 at the origin, a force F12 is experienced by Q2 due to electric field intensity E12 of Q1 at Q2

Coulomb's Law

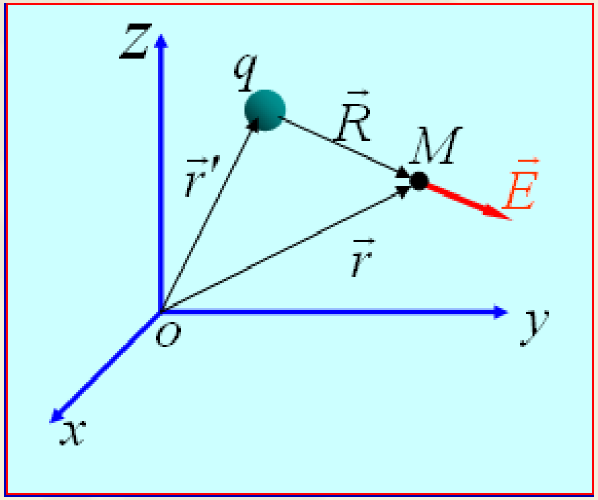
$$\boxed{\bar{F}_{12} = Q_2\bar{E}_{12} = \hat{a}_R \frac{Q_1 Q_2}{4\pi\varepsilon_0 R^2} \quad (N)}$$

• The permittivity of air is essentially the same as that of the free space:

$$\varepsilon_0 = 8.854187817\cdots\times 10^{-12}\,\text{F/m} \approx \frac{1}{36\pi}\times 10^{-9}\,\text{F/m}$$

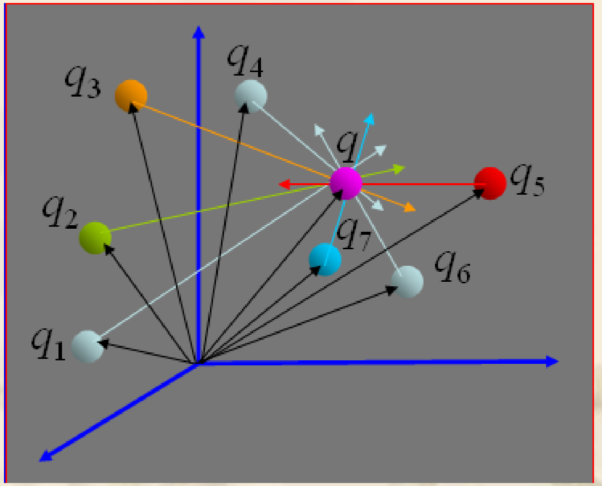
- **If a single point charge, Q, sits in an arbitrary location of a boundless free space, its E field would be:**

$$\vec{E}(\vec{r}) = \frac{q\vec{R}}{4\pi\varepsilon_0 R^3} \qquad \left(\vec{R} = \vec{r} - \vec{r}\,'\right)$$



- **Electric field created by a group of n discrete point charges q_1, q_2,… q_n, The principal of superposition applies:**

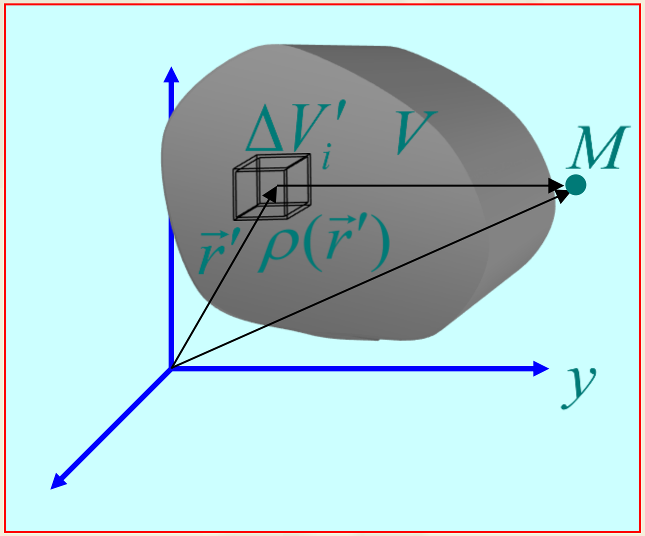
$$\vec{E} = \sum_{i=1}^{N} \vec{E}_{q_i} = \sum_{i=1}^{N} \frac{q_i \vec{R}_i}{4\pi\varepsilon_0 R_i^3}$$



Q: What if the charges are continuous distributed?

- Electric field intensity generated by volume distribution charges whose volume density is $\rho(\vec{r}')$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\vec{r}')\vec{R}}{R^3}\mathrm{d}V'$$



- Electric field intensity generated by surface distribution charges whose surface density is $\rho_s(\vec{r}')$

$$\vec{E}(r) = \frac{1}{4\pi\varepsilon_0}\int_S \frac{\rho_S(\vec{r}')\vec{R}}{R^3}\mathrm{d}S'$$

- Electric field intensity generated by line distribution charges whose line density is $\rho_l(\vec{r}')$

$$\vec{E}(r) = \frac{1}{4\pi\varepsilon_0}\int_C \frac{\rho_l(\vec{r}')\vec{R}}{R^3}\mathrm{d}l'$$

❖ Electric field intensities of several typical charge distributions

- **Electric field intensity of uniformly charged straight line segment :**

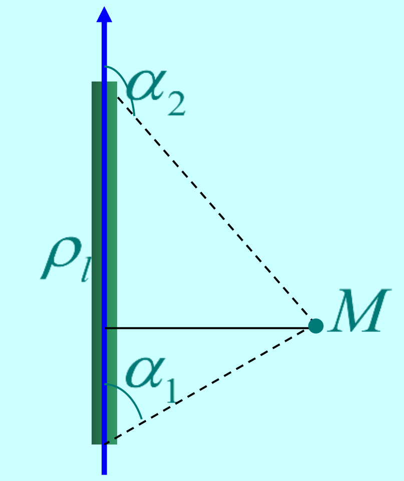
$$E_r = \frac{\rho_l}{4\pi\varepsilon_0 r}(\cos \quad _1 \quad \cos \quad _2)$$

$$E_z = \frac{\rho_l}{4\pi\varepsilon_0 r}(\sin \quad _2 \quad \sin \quad _1)$$

(finite)

$$E_r = \frac{\rho_l}{2\pi\varepsilon_0 r}$$

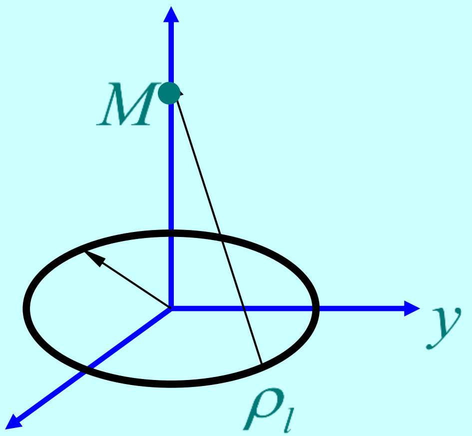
(infinite)



uniformly charged straight line segment

- **Electric field intensity of uniformly charged ring :**

$$E_z(0,0,z) = \frac{a\rho_l z}{2\varepsilon_0(a^2 \quad z^2)^{3/2}}$$

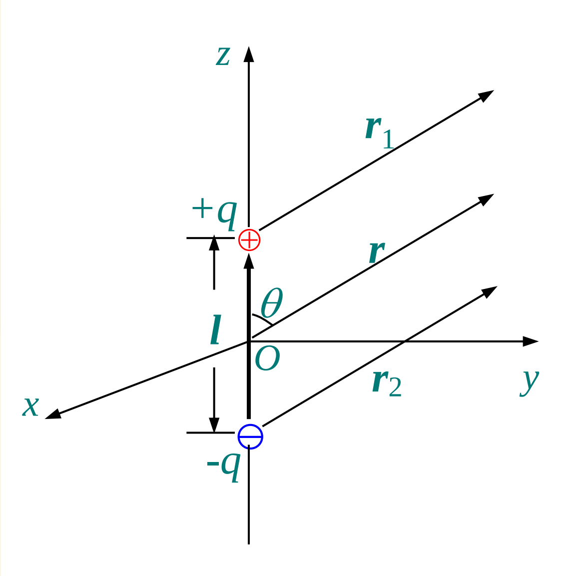


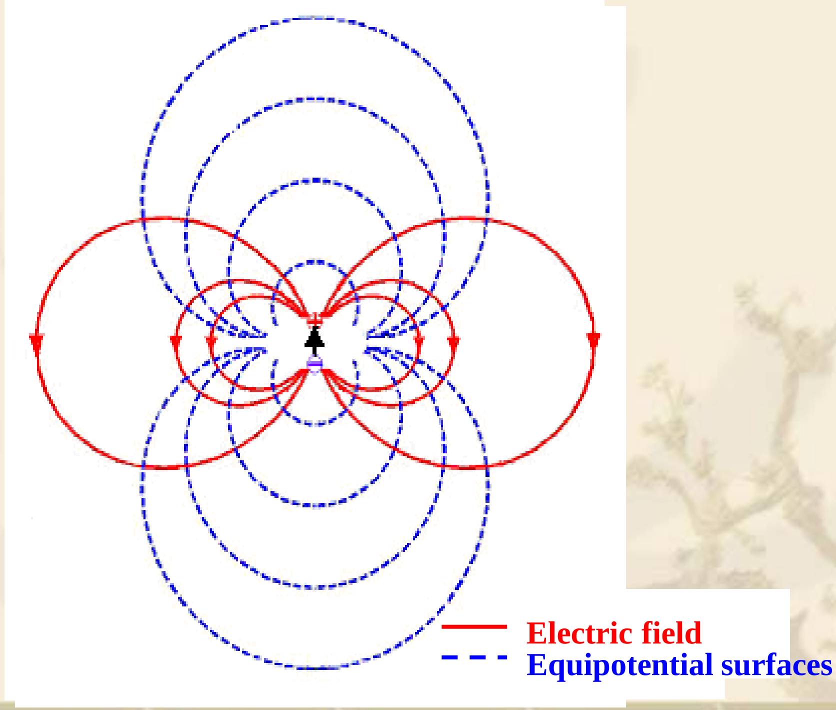
uniformly charged straight ring

The electric field intensity in far field produced by an electric dipole.

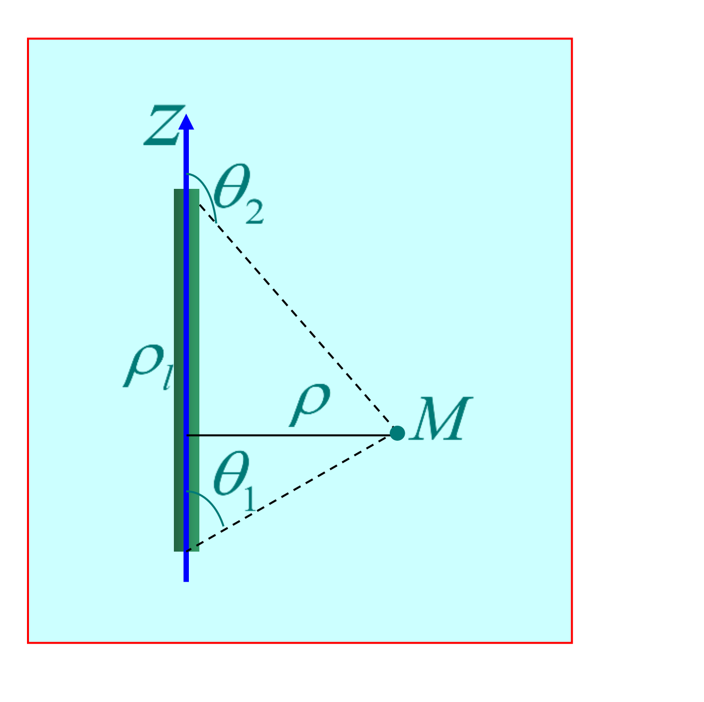
$$E = -\left(\boldsymbol{e}_r \frac{\partial \varphi}{\partial r} + \boldsymbol{e}_\theta \frac{1}{r} \frac{\partial \varphi}{\partial \theta} + \boldsymbol{e}_\phi \frac{1}{r \sin\theta} \frac{\partial \varphi}{\partial \phi} \right) = \boldsymbol{e}_r \frac{p \cos\theta}{2\pi\varepsilon_0 r^3} + \boldsymbol{e}_\theta \frac{p \sin\theta}{4\pi\varepsilon_0 r^3}$$

where $\boldsymbol{p} = q\boldsymbol{l}$ (**electric moment**), and thedirection of $\boldsymbol{l}$ is from negtive charge to the positive charge





Example 1 Find the electric field intensity caused by a line charge with length L and charge density ρ_l .



Solution: Let the z-axis of a cylindrical coordinate system coincide with the line charge and the mid point of the line be at the origin.

Since it is rotationally symmetrical, the field is independent of the angle ϕ .

It is impossible to apply Gauss' law to calculate the field because the direction of the electric field intensity cannot be found out in advance. We have to evaluate directly the electric potential and the electric field intensity.

Establish the Cartesian Coordinates so that field point M is on the surface passing through the origin point and perpendicular to z-axis

$$\boldsymbol{E} = \frac{\rho_l}{4\pi\varepsilon_0}\int_l \frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3}\mathrm{dz}'$$

Considering

$$z' = -r\frac{\cos\theta}{\sin\theta}$$

We have

$$\boldsymbol{E} = \frac{\rho_l}{4\pi\varepsilon_0}\left[\int_{z_1}^{z_2}\frac{\mathbf{e}_r r}{\left(r^2+z'^2\right)^{3/2}}\,\mathrm{dz}' - \int_{z_1}^{z_2}\frac{\mathbf{e}_z z'}{\left(r^2+z'^2\right)^{3/2}}\,\mathrm{dz}'\right]$$

$$= \frac{\rho_l}{4\pi\varepsilon_0 r}[(\sin\theta_2-\sin\theta_1)\mathbf{e}_z+(\cos\theta_1-\cos\theta_2)\mathbf{e}_r]$$

$$\mathbf{E} = \frac{\rho_l}{4\pi\varepsilon_0 r}[(\sin\theta_2 - \sin\theta_1)\mathbf{e}_z + (\cos\theta_1 - \cos\theta_2)\mathbf{e}_r]$$

If the length $L \to \infty$, then $\theta_1 \to 0$, $\theta_2 \to \pi$ and

$$\boldsymbol{E} = \frac{\rho_l}{4\pi\varepsilon_0 r} 2\boldsymbol{e}_r = \frac{\rho_l}{2\pi\varepsilon_0 r}\boldsymbol{e}_r$$

This result is the same as that of Example 3.

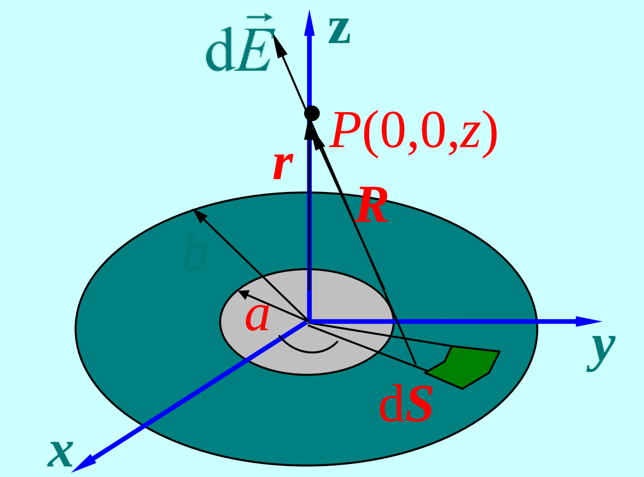
Example 2 **Calculate the electric field intensity at arbitrary point on the axis of uniformly charged circular thin disc.**

$$\vec{E}(r') = \frac{1}{4\pi\varepsilon_0}\int_S \frac{\rho_S(\vec{r}')\vec{R}}{R^3}\,\mathrm{d}S'$$

Field point: $P(0,0,z)$

Source point: $\vec{r}' = \vec{e}_\rho \rho'$ $\quad \vec{r} = \vec{e}_z z$

$$\vec{R} = \vec{r} - \vec{r}' \qquad \mathrm{d}S' = \rho'\mathrm{d}\rho'\mathrm{d}\phi'$$



uniformly charged circular thin ring

$$\vec{E}(\vec{r}) = \frac{\rho_S}{4\pi\varepsilon_0}\int_a^b\int_0^{2\pi}\frac{\vec{e}_z z - \vec{e}_\rho \rho'}{(z^2+\rho'^2)^{3/2}}\rho'\mathrm{d}\rho'\mathrm{d}\phi'$$

since $\int_0^{2\pi}\vec{e}_\rho\,\mathrm{d}\phi' = \int_0^{2\pi}(\vec{e}_x\cos\phi' + \vec{e}_y\sin\phi')\mathrm{d}\phi' = 0$

$$\vec{E}(\vec{r}) = \vec{e}_z\frac{\rho_S z}{2\varepsilon_0}\int_a^b\frac{\rho'\mathrm{d}\rho'}{(z^2+\rho'^2)^{3/2}} = \vec{e}_z\frac{\rho_S z}{2\varepsilon_0}\left[\frac{1}{(z^2+a^2)^{1/2}} - \frac{1}{(z^2+b^2)^{1/2}}\right]$$

3.4 Gauss's Law and Applications

- **Gauss' law follows directly from the divergence postulate of electrostatics by the application of divergence theorem.**

$$\oint_S \bar{E} \cdot d\bar{s} = \frac{Q}{\varepsilon_0}$$

• Gauss's law asserts that the total outward flux of the E-field over any closed surface in free space is equal to the total charge enclosed in the surface divided by ε_0 .

• Gauss's law is particularly useful in determining the E field of charge distributions with some symmetry conditions. On the other hand, when symmetry conditions do not exist, Gauss's law would not be of much help.

The condition of using Gauss's law to calculate the electric field strength

$$\oint_S \vec{E}(\vec{r}) \cdot \mathrm{d}\vec{S} = \frac{1}{\varepsilon_0} q\Big|_{inside\quad S}$$

Condition for simple calculation: $\vec{E}(\vec{r})$ can be moved outside from the integral sign, simplified the integral equation to algebra equation.

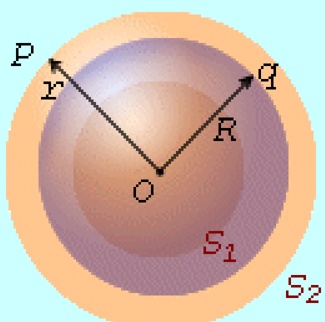
Under which circumstances, $\vec{E}(\vec{r})$ can be move to outside the integral sign?

$\vec{E}(\vec{r})$ **is uniform distribution on the S! Or integral result is known!**

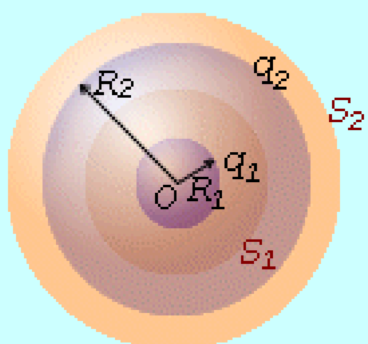
Where can we find this problem?

When the problem is symmetric!

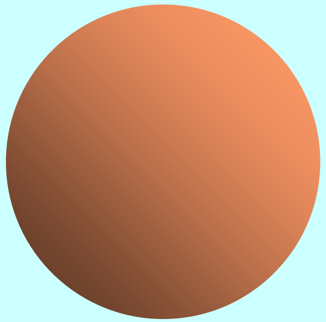
- **Spherical symmetric distribution: Including uniformly charged spherical surface, sphere and multi layer concentric spherical shell etc.**



Charged spherical shell

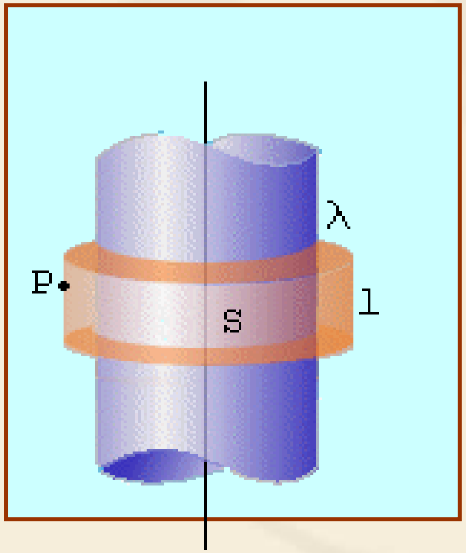


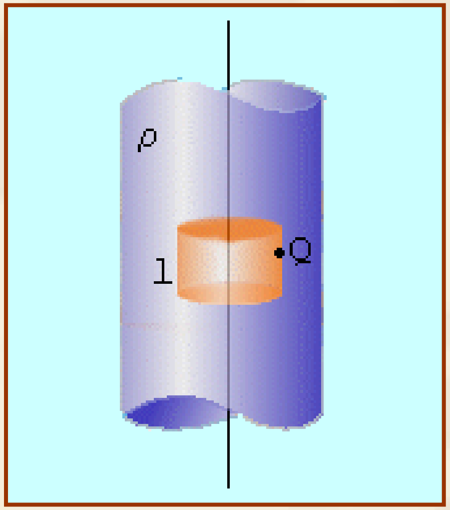
Multi-layer concentric Spherical shell



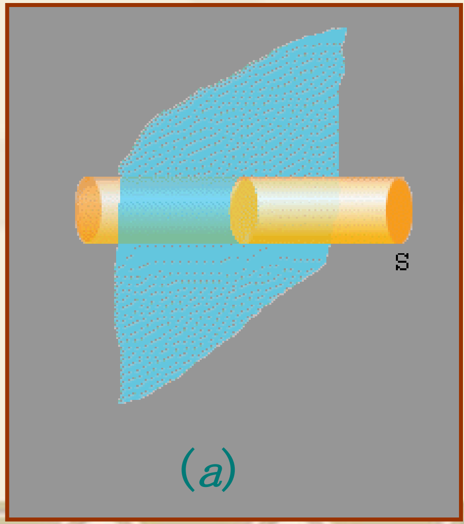
Uniformly charged sphere

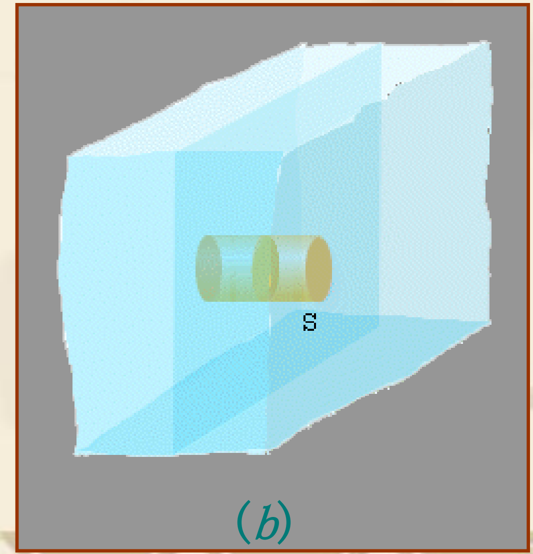
•Axial symmetrical distribution： Such as infinite uniformly charged straight line 、 Cylindrical surface 、 cylindrical shell, etc.





•Infinite plane charge： Such as infinitely large uniform charged plane 、flat, etc.





Example 3 **Solving the electric field generated by a uniformly charged sphere in vacuum . Known sphere radius is a， and the charge density is ρ_0。**

Solution：（1）Field strength of a certain point outside the sphere

$$\oint_S \vec{E}\cdot d\vec{S} = \frac{q}{\varepsilon_0} = \frac{1}{\varepsilon_0}\frac{4}{3}\pi a^3 \rho_0$$

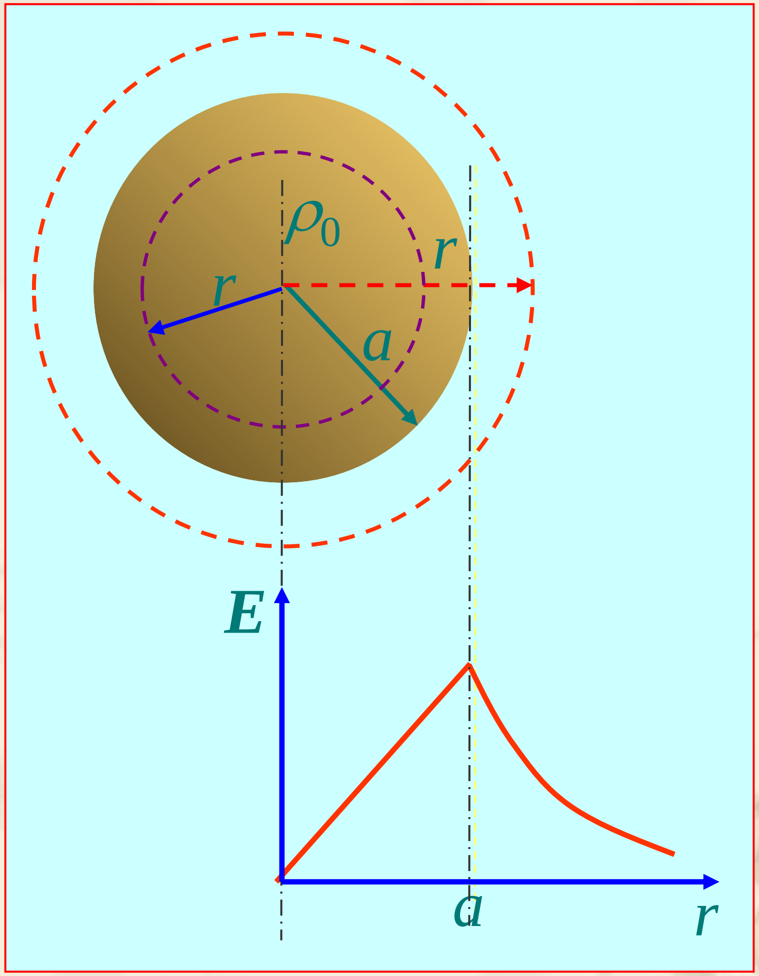
$$\Rightarrow E = \frac{\rho_0 a^3}{3\varepsilon_0 r^2} \qquad (r \geq a)$$

（2）Field strength of a certain point inside the sphere

$$\oint_S \vec{E}\cdot d\vec{S} = \frac{1}{\varepsilon_0}\int_V \rho_0 dV$$

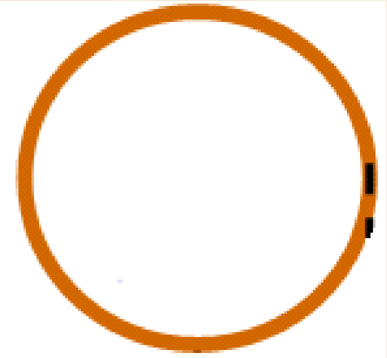
$$\Rightarrow 4\pi r^2 E = \frac{1}{\varepsilon_0}\cdot\frac{q}{4\pi a^3/3}\cdot\frac{4}{3}\pi r^3$$

$$\Rightarrow E = \frac{\rho_0 r}{3\varepsilon_0} \qquad (r < a)$$



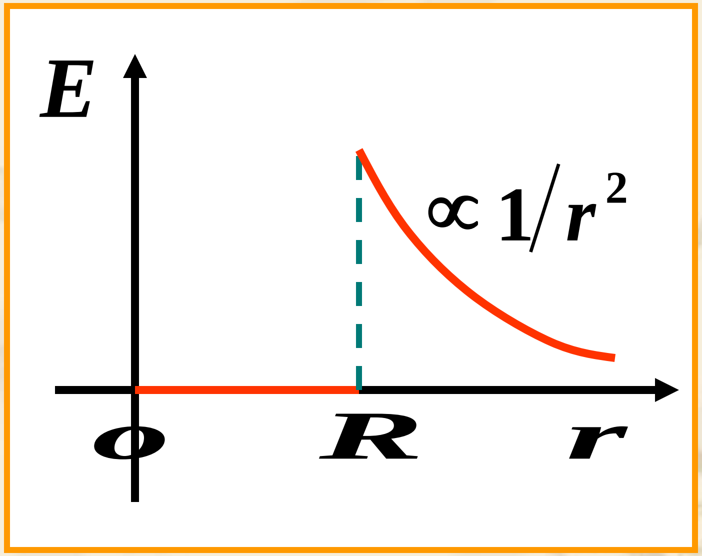
Discussion

- **Find the electric field generated by a uniformly charged surface in vacuum**

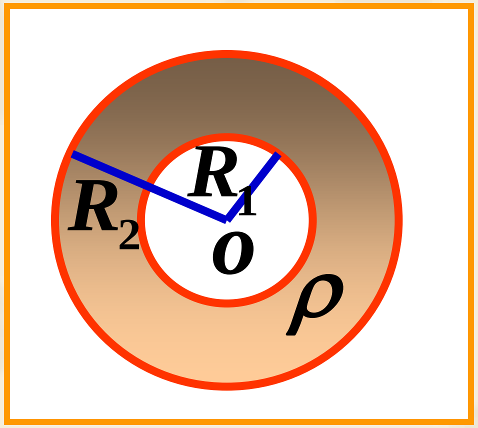


Gaussian surface: **Concentric spherical surface with radius of r**

$$\vec{E} = \begin{cases} 0 & (r < R) \\ \dfrac{q\vec{r}}{4\pi\varepsilon_0 r^3} & (r > R) \end{cases}$$



- **Understand the discontinuity of E-field on the surface r=R**

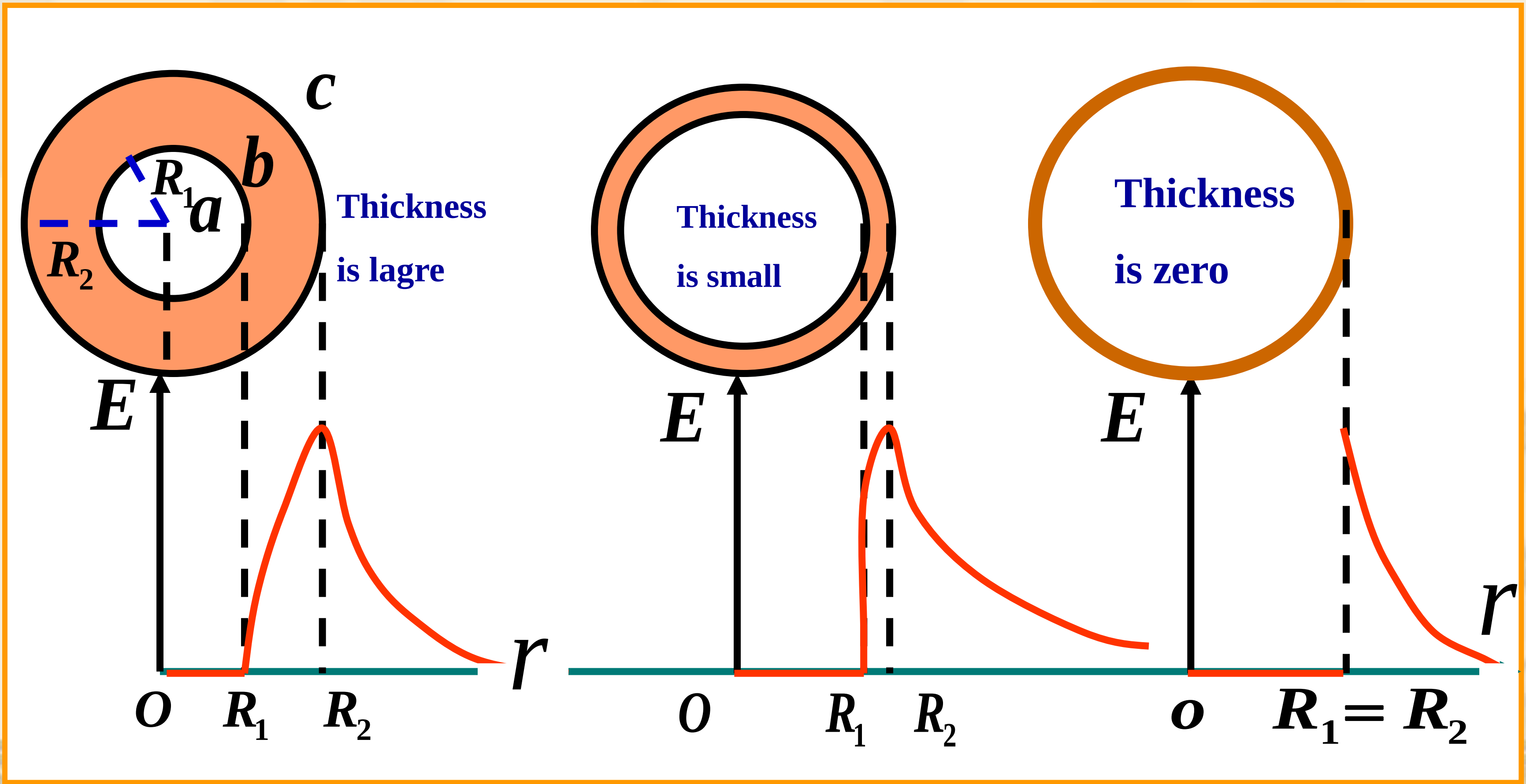


Find the E-field with $R_1, \quad R_2, \quad \rho$

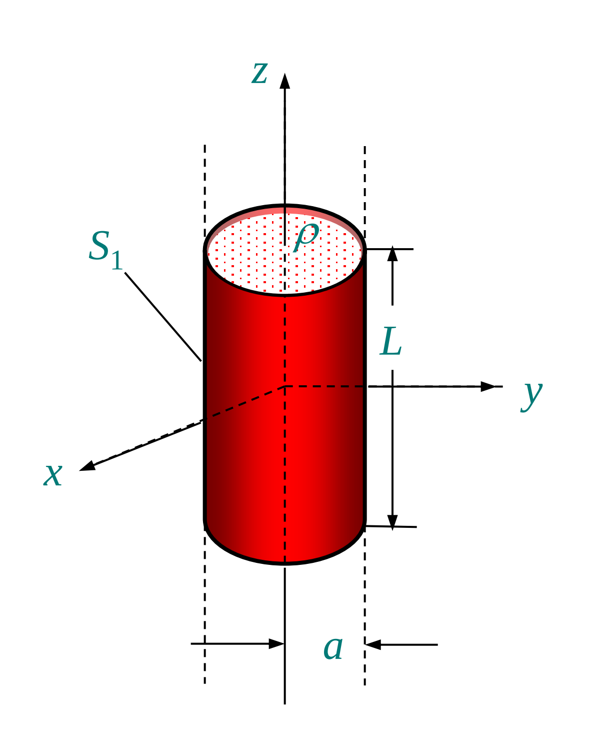
for the uniformly charged shell

$$E = \begin{cases} 0 & (r \le R_1) \\ \dfrac{\rho}{3\varepsilon_0}\left(r - \dfrac{R_1^3}{r^2}\right) & (R_1 \le r \le R_2) \\ \dfrac{\rho(R_2^3 - R_1^3)}{3\varepsilon_0 r^2} = \dfrac{q}{4\pi\varepsilon_0 r^2} & (r \ge R_2) \end{cases}$$

- **Applying surface model** is the reason for the discontinuity of E-field on the surface. For real application, **the volume charge density** should be applied for the accurate calculation.



Example 1 Assume that an infinitely long charged cylinder of radius *a* is placed in free space. The density of the charge is ρ . Calculate the electric field intensities inside and outside the cylinder.



Solution: Choose a cylindrical coordinate system. Since the cylinder is infinitely long, for any *z* the environment remain the same. Hence the field is independent of the coordinate variable *z*.

Since it is symmetrical about any plane of *z* = constant, the electric field intensity must be perpendicular to the *z*-axis, and coplanar with the radial coordinate *r*.

In addition, the cylinder is rotationally symmetrical. Thus the field must be independent of the angle ϕ .

Construct a cylindrical **Gaussian surface** of radius ***r*** and length ***L***. Applying Gauss' law, we have

$$\oint_S \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = \frac{q}{\varepsilon_0}$$

Since the direction of the electric field intensity **coincides** with the outward normal of lateral surface S_1 of the cylinder **everywhere** and is **perpendicular** to the normal to the upper or the lower end faces, the left side of the surface integral becomes

$$\oint_S \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = \int_{S_1} E\mathrm{d}S = E\int_{S_1} \mathrm{d}S = 2\pi rLE$$

For $r < a$, the charge $q = \pi r^2 \rho L$, and the electric field intensity is

$$\boldsymbol{E} = \frac{\rho\, r}{2\varepsilon_0}\boldsymbol{e}_r$$

When $r > a$, the charge $q = \pi a^2 \rho L$, we have

$$\boldsymbol{E} = \frac{\pi a^2 \rho}{2\pi\varepsilon_0 r}\boldsymbol{e}_r$$

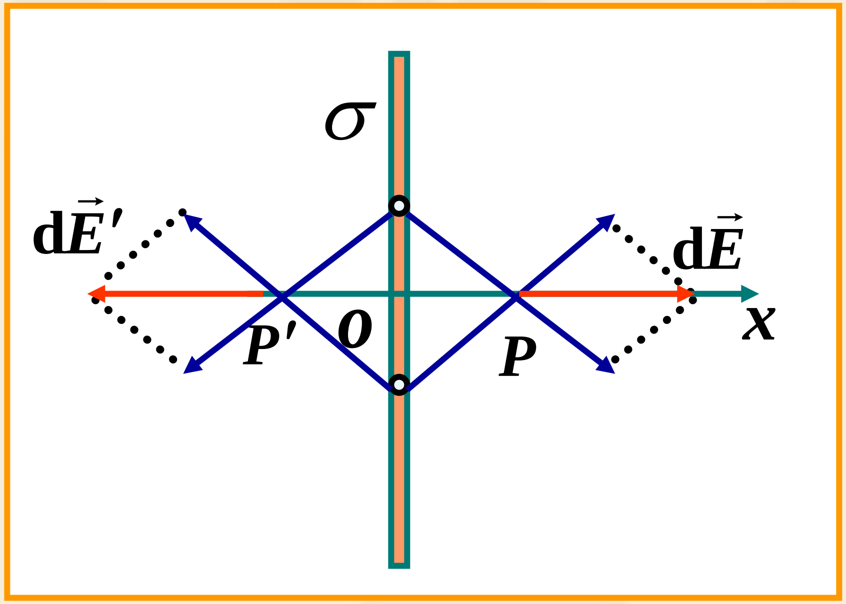
where $\pi a^2 \rho$ can be considered as the charge per unit length, so that the electric field can be thought of as caused by a line charge with density $\rho_l = \pi a^2 \rho$. In view of this we can derive the electric field intensity caused by an infinite line charge with line density ρ_l as

$$\boldsymbol{E} = \frac{\rho_l}{2\pi\varepsilon_0 r}\boldsymbol{e}_r$$

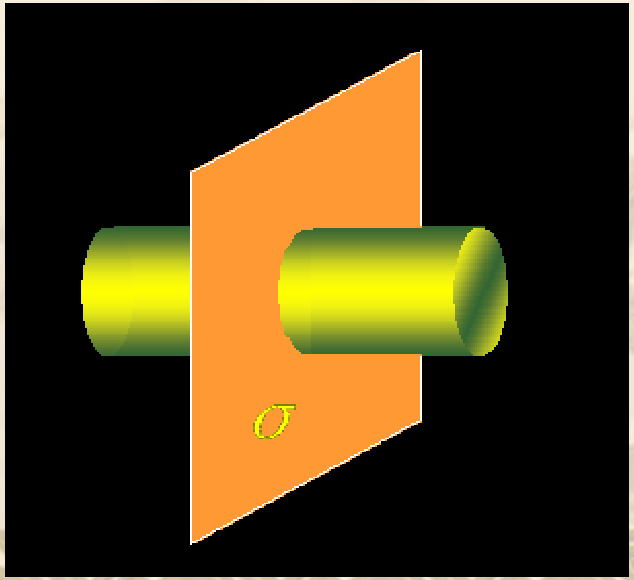
For this example, it is easy to calculate the electric field intensity using Gauss' law. If we directly calculate the electric field intensity from the distribution of the charge, it will be very complicated.

Example 7 **Determine the electric field intensity of an infinite planar charge with a uniform surface charge density σ 。**

The collection of infinite uniform charged straight line



$\vec{E}$ is normal to the infinite planar



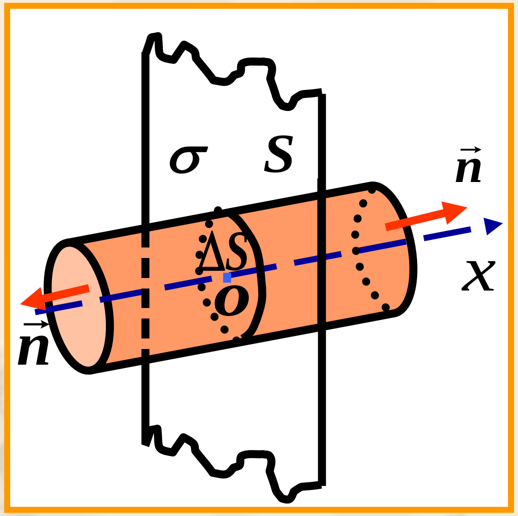
How to build the Gaussian surface?

Build the Gaussian surface

$$\oint_S \vec{E}\cdot d\vec{S} = \int_{\text{left}} \vec{E}\cdot d\vec{S} + \int_{\text{right}} \vec{E}\cdot d\vec{S} + \int_{\text{side}} \vec{E}\cdot d\vec{S}$$

$$= \int_{\text{left}} E\cos 0^\circ dS + \int_{\text{right}} E\cos 0^\circ dS + \int_{\text{side}} E\cos\frac{\pi}{2} dS$$

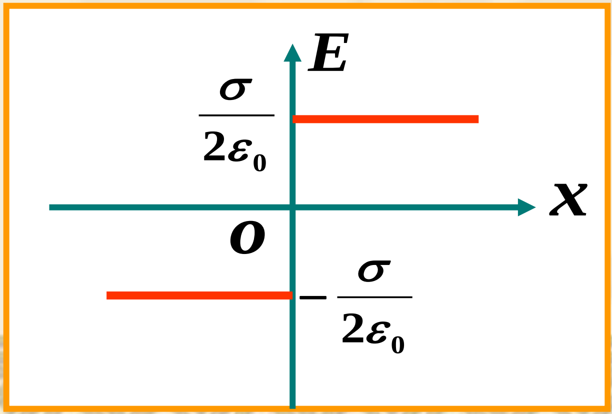
$$= E\cdot 2\Delta S$$



Gauss's Law:

$$\oint \vec{E}\cdot d\vec{S} = E\cdot 2\Delta S$$

$$= \frac{1}{\varepsilon_0}\sum q_{\text{inside}} = \frac{\sigma\Delta S}{\varepsilon_0}$$



$$E = \frac{\sigma}{2\varepsilon_0}$$

The direction of E depends on the sign of the charge

3.5 Electric Potential

- **Electric Potential (scalar field)**

since $\nabla\times(\nabla\varphi)\equiv 0$ **and** $\nabla\times\vec{E}=0$

We can define a scalar electric potential φ such that

$$\Rightarrow \quad \vec{E}=-\nabla\varphi$$

Advantage： change the vector problem into scalar problem

The physical meaning of the electric potential φ at a point is the amount of work needed to move a unit of positive charge from a reference point to a specific point inside the field without producing an acceleration., Hence

$$\varphi=\frac{W}{q}$$

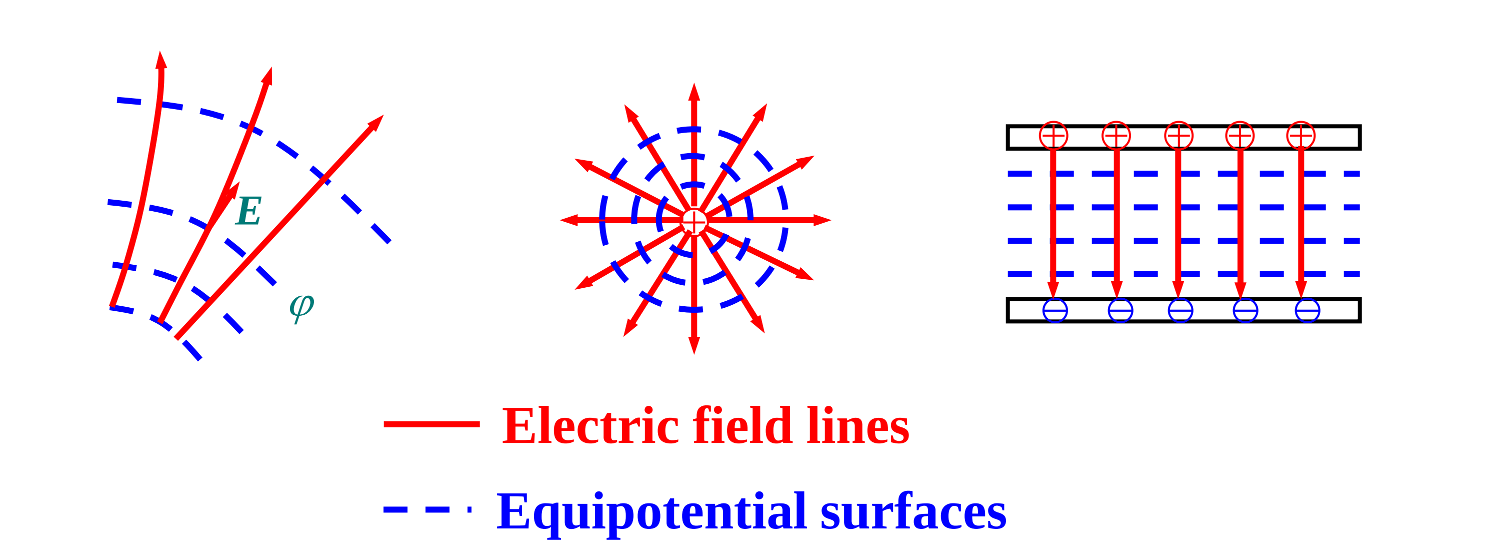
(q is the amount of the charge)

❖ Equipotential surfaces

Equipotential surfaces are the surfaces on which the electric potential are equalized, and the equation is $\varphi(x, y, z) = C$

➢ **The electric field lines are perpendicular to the equipotential surfaces everywhere.**

➢ **The distribution density of equipotential surfaces can indicate the intensity of the electric field.**



The relationship between E and φ

- **Find φ** $\quad \mathrm{d}\varphi = \nabla\varphi \cdot \mathrm{d}\vec{l} = -\vec{E}\cdot \mathrm{d}\vec{l}$ $\Rightarrow$ $\varphi = -\int \vec{E}\cdot \mathrm{d}\vec{l}$
- **Find E** $\quad \vec{E} = -\nabla\varphi$

The differential equation of φ

In the homogeneous medium

$$\left.\begin{aligned} \nabla\cdot\vec{E} &= \rho/\varepsilon_0 \\ \vec{E} &= -\nabla\varphi \end{aligned}\right\}$$

$\vec{\mathbf{E}}_1 = -\nabla\varphi_1$

$\nabla^2\varphi_1 = -\rho/\varepsilon$

Charge

Sub−1 ε_1 $\quad \nabla^2\varphi_2 = 0$

Sub−2 ε_2 $\quad \nabla^2\varphi_1 = 0$

$\vec{\mathbf{E}}_2 = -\nabla\varphi_2$

Scalar poisson equation

$$\Rightarrow \nabla^2\varphi = -\rho/\varepsilon$$

When, $\rho = 0$

$$\Rightarrow \nabla^2\varphi = 0$$

Laplace equation

- Solve the problem in homogeneous medium using electric potential

With point charge

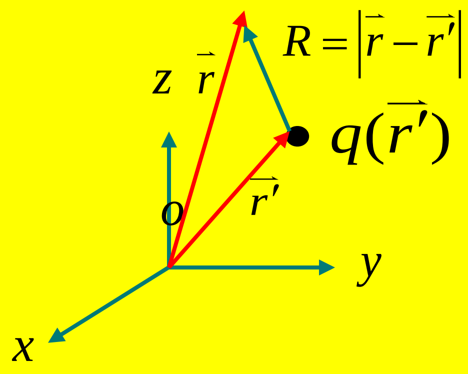
- **Point charge**: volume→0; density →∞, volume·density=1
- **Delta function**: the density of unit point charge

$$\nabla^2 \varphi(\vec{r}) = -\frac{q}{\varepsilon}\delta(\vec{r}-\vec{r}\,')$$

$$\varphi(\vec{r}) = \frac{q}{4\pi\varepsilon R} + C$$

$$\vec{\mathbf{E}} = -\nabla\varphi$$

$$\nabla^2\varphi = 0$$



Sub ε

- **Solve the problem in homogeneous medium using electric potential**

➤ **With arbitrary charge source: (superposition)**

$$dq \Rightarrow d\varphi \qquad d\varphi(\vec{r}) = \frac{dq}{4\pi\varepsilon R}$$

- **Volume charge source**

$$dq = \rho dv' \longrightarrow \varphi(\vec{r}) = \iiint \frac{\rho(\vec{r'})dv'}{4\pi\varepsilon R} + C$$

- **Surface charge source**

$$dq = \rho_s dS \longrightarrow \varphi(\vec{r}) = \iint \frac{\rho_s ds}{4\pi\varepsilon R} + C$$

- **Linear charge source**

$$dq = \rho_l dl \longrightarrow \varphi(\vec{r}) = \int \frac{\rho_l dl}{4\pi\varepsilon R} + C$$

Voltage - potential difference

$$\varphi = -\int \vec{E} \cdot \mathrm{d}\vec{l}$$

$$\varphi(P_1) - \varphi(P_2) = \int_{P_2}^{P_1} d\varphi = -\int_{P_2}^{P_1} E \cdot \mathrm{d}\vec{l} = \int_{P_1}^{P_2} E \cdot \mathrm{d}\vec{l}$$

Potential difference between P1 and P2

Work by the electric force

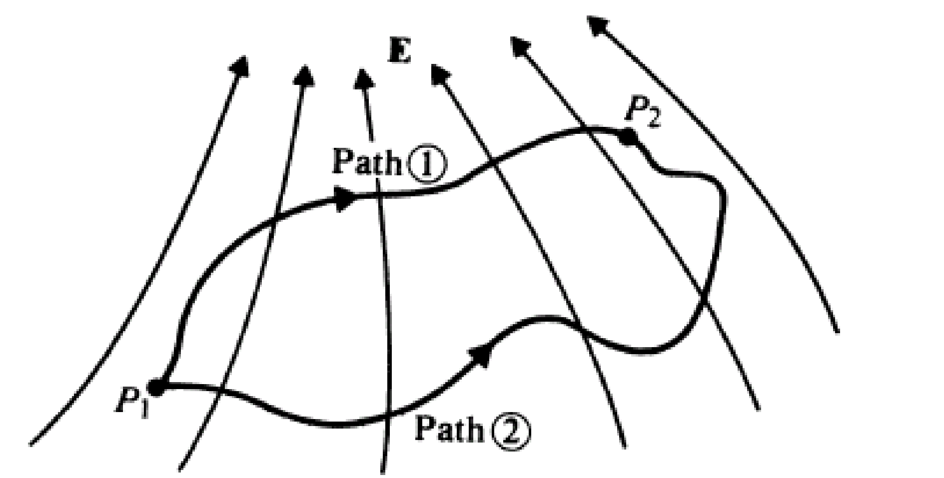


FIGURE 3–11
Two paths leading from P_1 to P_2 in an electric field.

Q: Choosing different integral path will change the result of voltage?

❖ Potential reference point

- **Electrostatic potential is not unique, could be different with a constant value;**
- **To determine the potential of a point by choosing a reference point**
- **The potential difference between any two points is independent of the reference point**

Choose reference point → **Set the reference point as zero potential** → **Determine the potential**

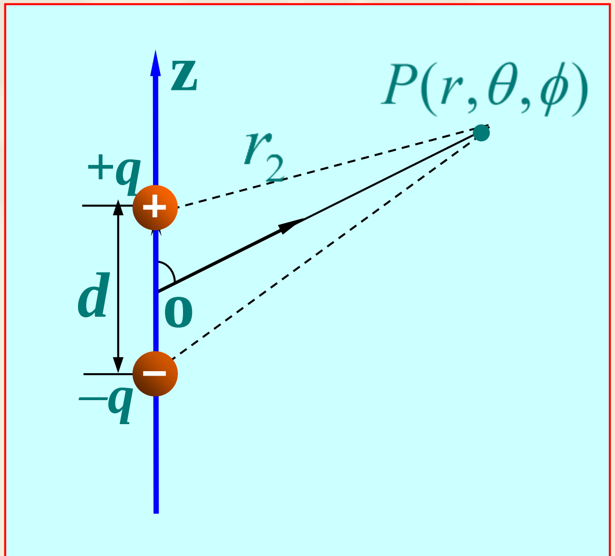
The principle to choose a reference point

The potential expressions is meaningful;

The potential expressions is the simplest;

Only one reference point for one problem.

Example 8 **Find the potential and electric field of an electric dipole.**



Electric dipole

Solution: In the spherical system

$$\varphi(\vec{r}) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = \frac{q}{4\pi\varepsilon_0}\frac{r_2 - r_1}{r_1 r_2}$$

$$r_1 = \sqrt{r^2 + (d/2)^2 - rd\cos\theta}$$

$$r_2 = \sqrt{r^2 + (d/2)^2 + rd\cos\theta}$$

Binomial expansion, $r >> d$, **so** $r_1 = r - \frac{d}{2}\cos\theta$, $r_2 = r + \frac{d}{2}\cos\theta$

So $$\varphi(\vec{r}) = \frac{qd\cos\theta}{4\pi\varepsilon_0 r^2} = \frac{\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{e}}_r}{4\pi\varepsilon_0 r^2} = \frac{\vec{\boldsymbol{p}}\cdot\vec{r}}{4\pi\varepsilon_0 r^3}$$

$\vec{p} = q\vec{d}$ **is the dipole moment, directed from the negative charge to positive charge**

$$\vec{E}(\vec{r}) = -\nabla\varphi = -(\vec{e}_r\frac{\partial\varphi}{\partial r} + \vec{e}_\theta\frac{1}{r}\frac{\partial\varphi}{\partial\theta} + \vec{e}_\phi\frac{1}{r\sin\theta}\frac{\partial\varphi}{\partial\phi})$$

$$= \frac{q}{4\pi\varepsilon_0 r^3}(\vec{\boldsymbol{e}}_r 2\cos\theta + \vec{\boldsymbol{e}}_\theta \sin\theta)$$

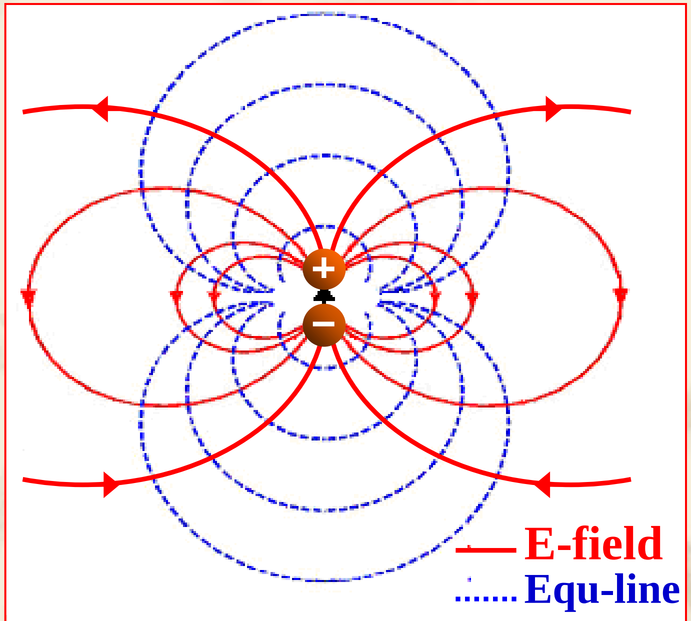
- **Equipotential line equation**:

$$\frac{p\cos\theta}{4\pi\varepsilon_0 r^2} = C \Rightarrow r^2 = C'\cos\theta$$

- **Differential equation of electric field line**:

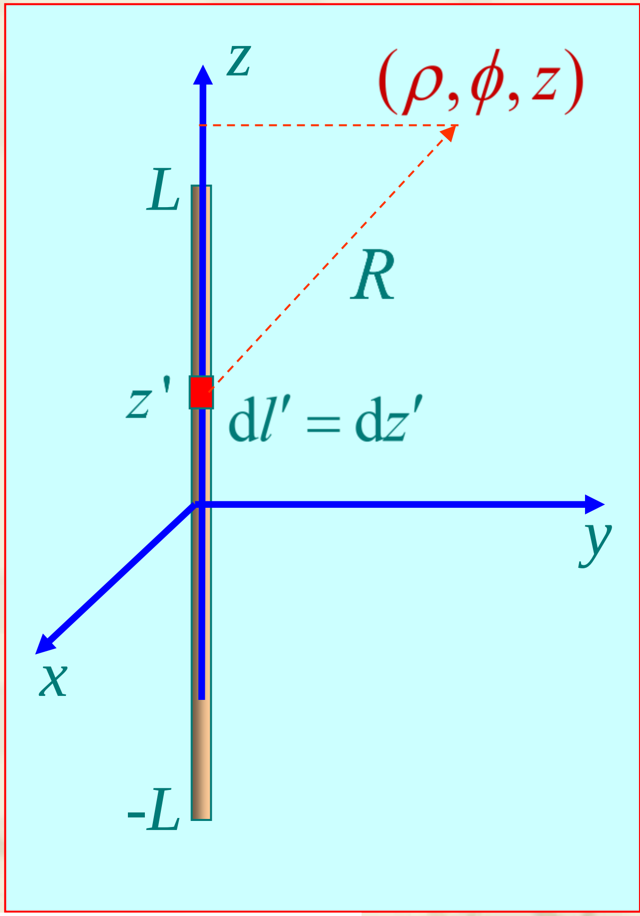
$$\frac{\mathrm{d}r}{E_r} = \frac{r\mathrm{d}\theta}{E_\theta}$$

So $$r = C_1\sin^2\theta$$



E-field for dipole

Example 9 **Find the potential and electric field of an uniformly charged line with the length of *2L* and charge density of ρ_{l0} .**



Solution: In the cylindrical system

$$\varphi(\vec{r}') = \frac{\rho_{l0}}{4\pi\varepsilon_0}\int_{-L}^{L}\frac{1}{\sqrt{\rho^2+(z-z')^2}}\mathrm{d}z' + C$$

$$= \frac{\rho_{l0}}{4\pi\varepsilon_0}\ln[z'-z+\sqrt{\rho^2+(z-z')^2}]\Big|_{-L}^{L} + C$$

$$= \frac{\rho_{l0}}{4\pi\varepsilon_0}\ln\frac{\sqrt{\rho^2+(z-L)^2}-(z-L)}{\sqrt{\rho^2+(z+L)^2}-(z+L)} + C$$

Choose a proper reference point to determine the C, for example $\varphi(\infty)=0$, then C=0

The electric potential produced by the infinite long line charge source

$$\varphi(\vec{r}\,) = \lim_{L\to\infty} \frac{\rho_{l0}}{4\pi\varepsilon_0} \ln \frac{\sqrt{\rho^2 + (z-L)^2} - (z-L)}{\sqrt{\rho^2 + (z+L)^2} - (z+L)} + C$$

$$= \frac{\rho_{l0}}{2\pi\varepsilon_0} \ln \frac{2L}{\rho} + C$$

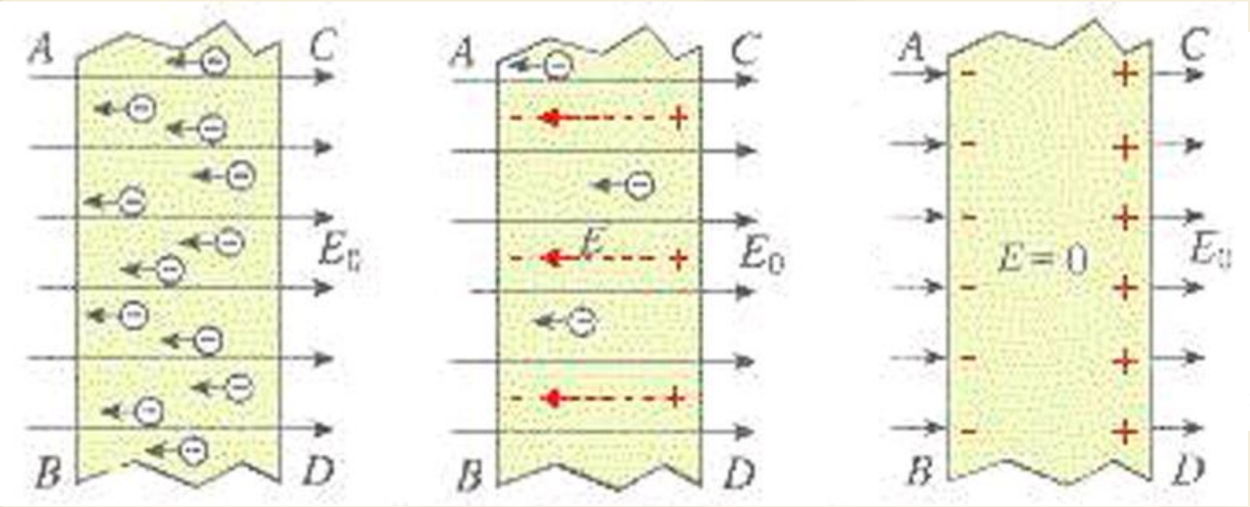
Choose a certain point, for example, $\rho = a$, then

$$C = -\frac{\rho_{l0}}{2\pi\varepsilon_0} \ln \frac{2L}{a} \Longrightarrow \varphi(\vec{r}\,) = \frac{\rho_{l0}}{2\pi\varepsilon_0} \ln \frac{a}{\rho}$$

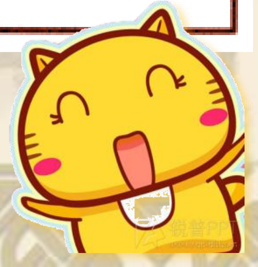
How to choose the most useful reference point?

3.6 Conductors in Static Electric Field

- Assume a conductor was stationed in a static electric field.
- The field will in the conductor exerting a force on the charges and making them move away from one another.
- This movement will continue until all the charges reach the conductor surface and redistribute themselves in such a way that both the charge and the field inside vanish.



Inside a Conductor (Under Static Conditions)
$\rho = 0$
$\mathbf{E} = 0$



- The tangential component of the electric field must be zero.

$$\oint_{abcda} \mathbf{E} \cdot d\ell = E_t \Delta w = 0 \qquad E_t = 0,$$

- E field on a conductor surface is everywhere normal to the surface.

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = E_n \Delta S = \frac{\rho_s \Delta S}{\epsilon_0} \qquad E_n = \frac{\rho_s}{\epsilon_0}.$$

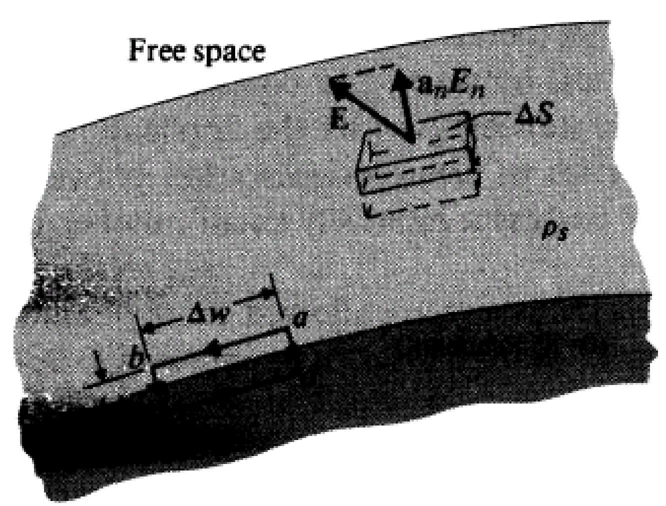
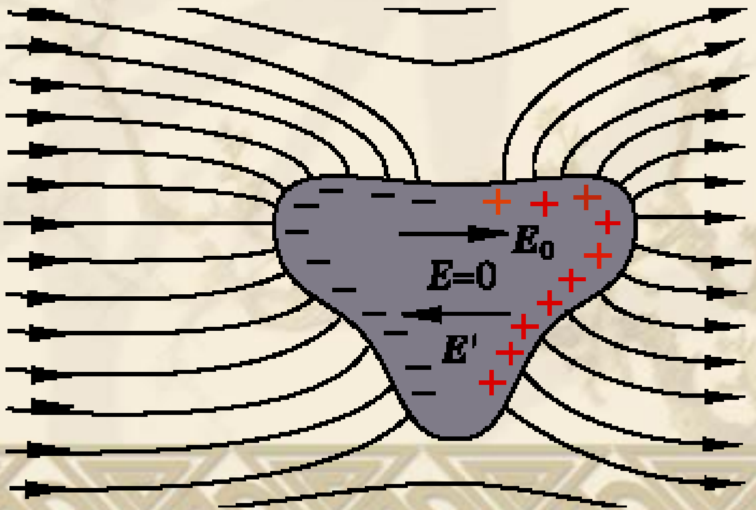


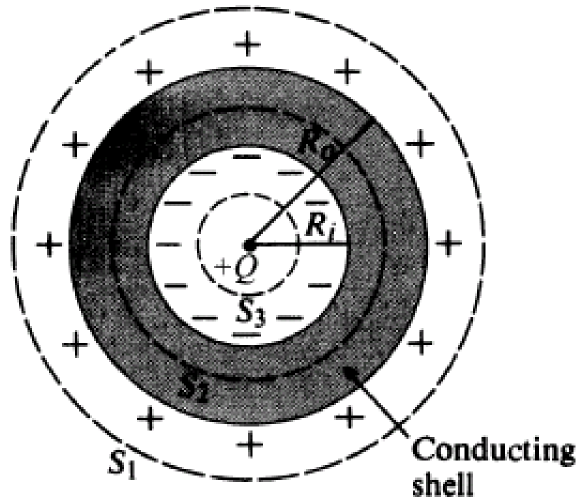
FIGURE 3–18
A conductor–free space interface.

Boundary Conditions at a Conductor/Free Space Interface
$E_t = 0$
$E_n = \frac{\rho_s}{\epsilon_0}$



The charge distribution on the surface of a conductor depends on the shape of the surface.

Example 10 **A positive point charge Q is at the center of a spherical conducting shell. Determine E and φ as functions of the radial distance R.**



(a)

a) $R > R_o$ (Gaussian surface S_1):

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = E_{R1} 4\pi R^2 = \frac{Q}{\epsilon_0}$$

or

$$E_{R1} = \frac{Q}{4\pi\epsilon_0 R^2}. \tag{3–73}$$

The **E** field is the same as that of a point charge Q without the presence of the shell. The potential referring to the point at infinity is

$$V_1 = -\int_{\infty}^{R} (E_{R1})\, dR = \frac{Q}{4\pi\epsilon_0 R}. \tag{3–74}$$

b) $R_i < R < R_o$ (Gaussian surface S_2): Because of Eq. (3–70), we know that

$$E_{R2} = 0. \tag{3–75}$$

Since $\rho = 0$ in the conducting shell and since the total charge enclosed in surface S_2 must be zero, an amount of negative charge equal to $-Q$ must be induced on the inner shell surface at $R = R_i$. (This also means that an amount of positive charge equal to $+Q$ is induced on the outer shell surface at $R = R_o$.) The conducting shell is an equipotential body. Hence,

$$V_2 = V_1\Big|_{R=R_o} = \frac{Q}{4\pi\epsilon_0 R_o}. \tag{3–76}$$

c) $R < R_i$ (Gaussian surface S_3): Application of Gauss's law yields the same formula for E_{R3} as E_{R1} in Eq. (3–73) for the first region:

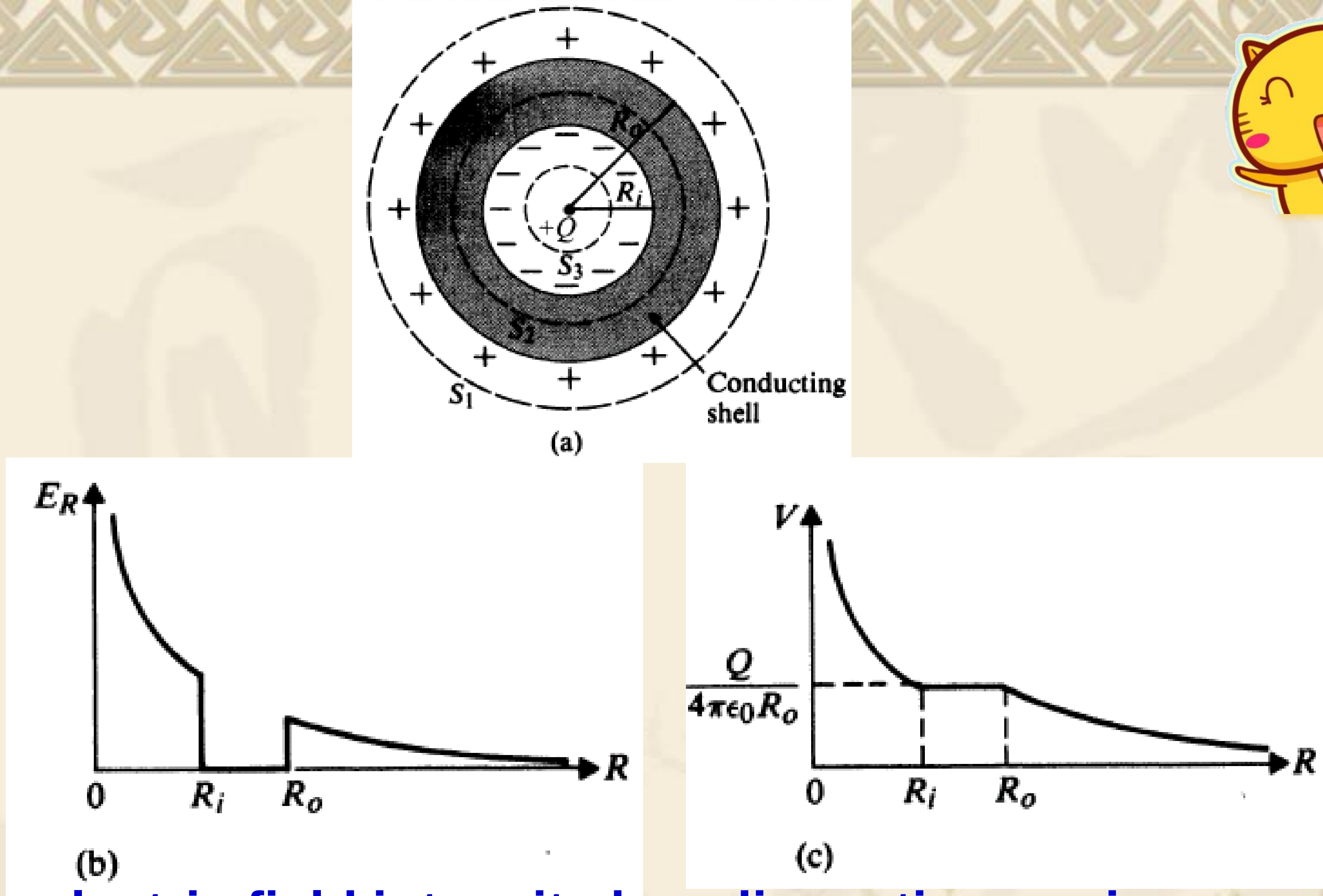
$$E_{R3} = \frac{Q}{4\pi\epsilon_0 R^2}. \tag{3–77}$$

The potential in this region is

$$V_3 = -\int E_{R3}\, dR + C = \frac{Q}{4\pi\epsilon_0 R} + C,$$

where the integration constant C is determined by requiring V_3 at $R = R_i$ to equal V_2 in Eq. (3–76). We have

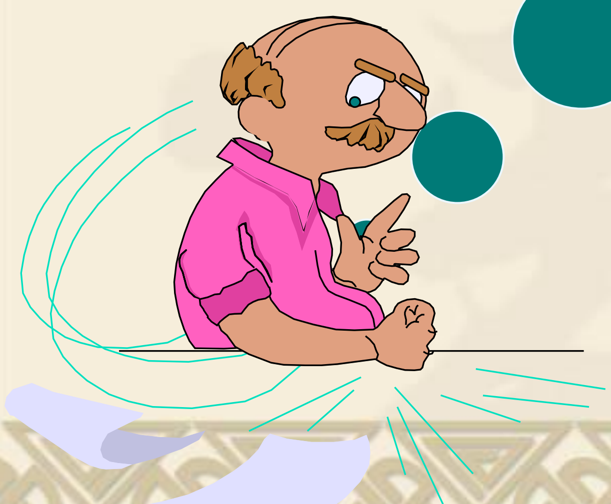
$$C = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{R_o} - \frac{1}{R_i}\right) \qquad V_3 = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{R} + \frac{1}{R_o} - \frac{1}{R_i}\right).$$



- The electric field intensity has discontinuous jumps, while the potential remains continuous;
- A discontinuous jump in potential would mean an infinite electric field intensity.

When medium exist in space：

- what will happen to the static electric fields produced by the charges?



3.7 Dielectrics in Static Electric Field

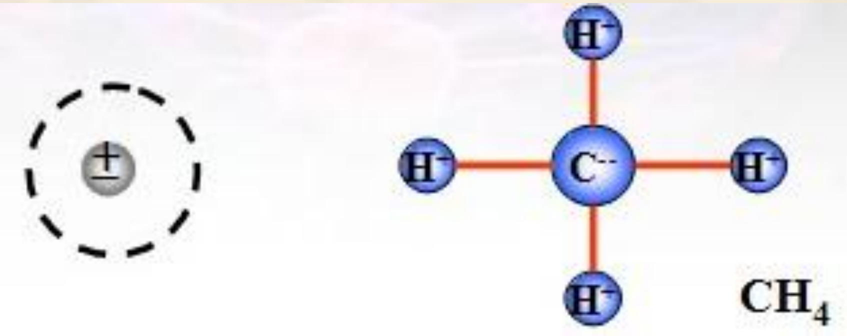
❖ **Three cases for the EM response：**

- **Polarization:** the bound charges in dielectric will be displaced due to the electric fields;
- **magnetization:** macroscopic magnetic moment results under the influence of an external magnetic field;
- **conduction:** the motion of the free electrons by an external E field.

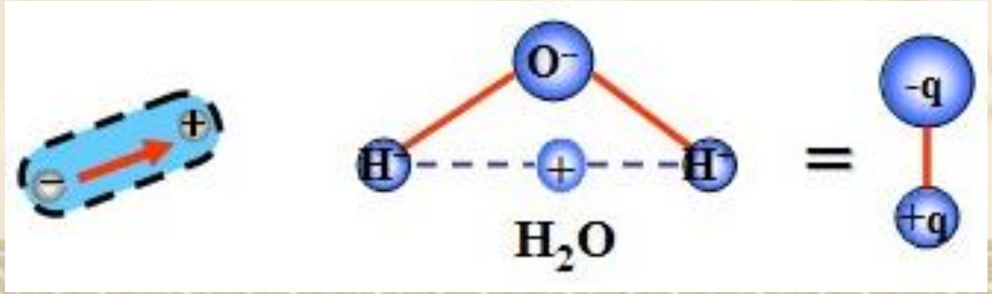
*** Parameters:**

- Dielectric constant (permittivity) ε
- Permeability μ
- conductivity σ

• **Nonpolar molecules:** **the centers of the positive and negative charges of an atom overlap each other, dipole moment will be zero. -** ***CO_2 ,H_2 , O_2 , CH_4***



• **Polar molecules:** **the centers of the positive and negative charges of an atom are separated, have a finite dipole moment. -** ***H_2O, HCl , CO, SO_2***



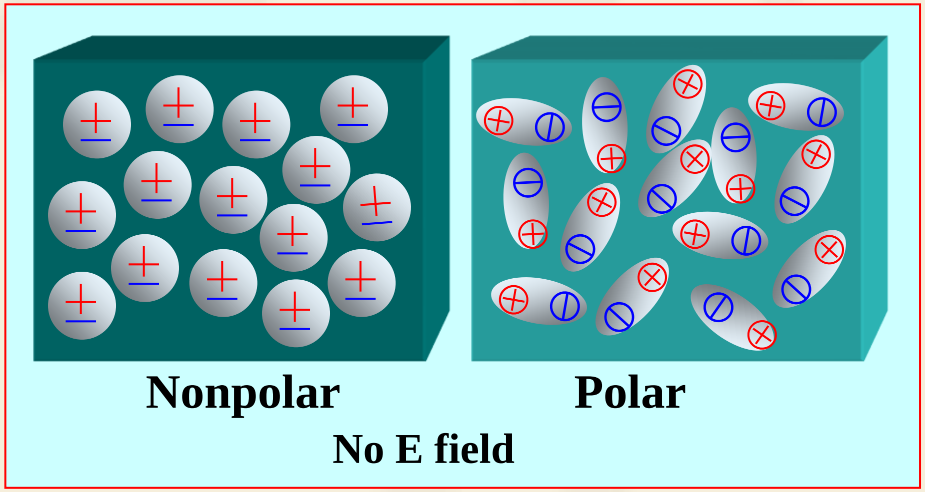
Polarization of dielectrics

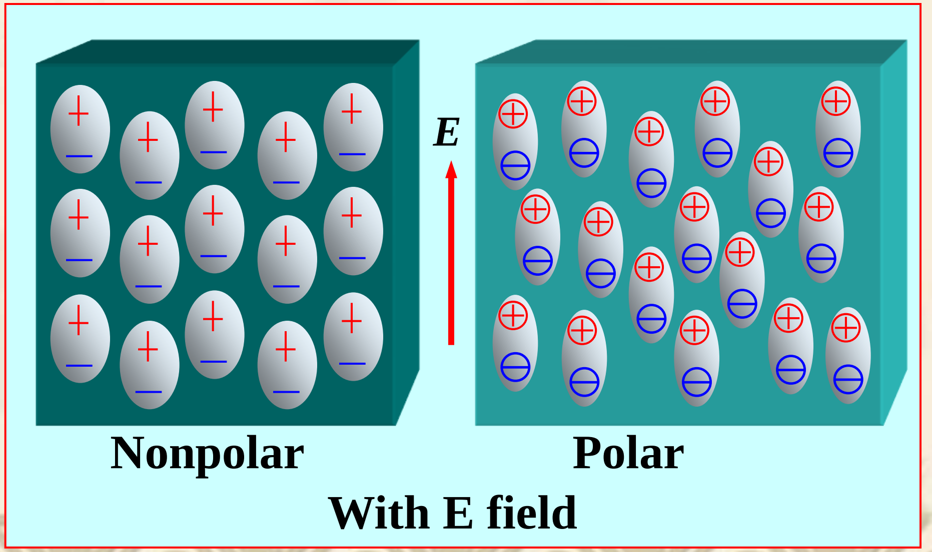
1. Phenomenon

1) With external E field, dielectric will have polarization :

- **Nonpolar molecules:**
  displacement polarization
- **Polar molecules:**
  orientation polarization

2) The magnitude of the polarization is determined by the number of electric dipole moments in the medium.

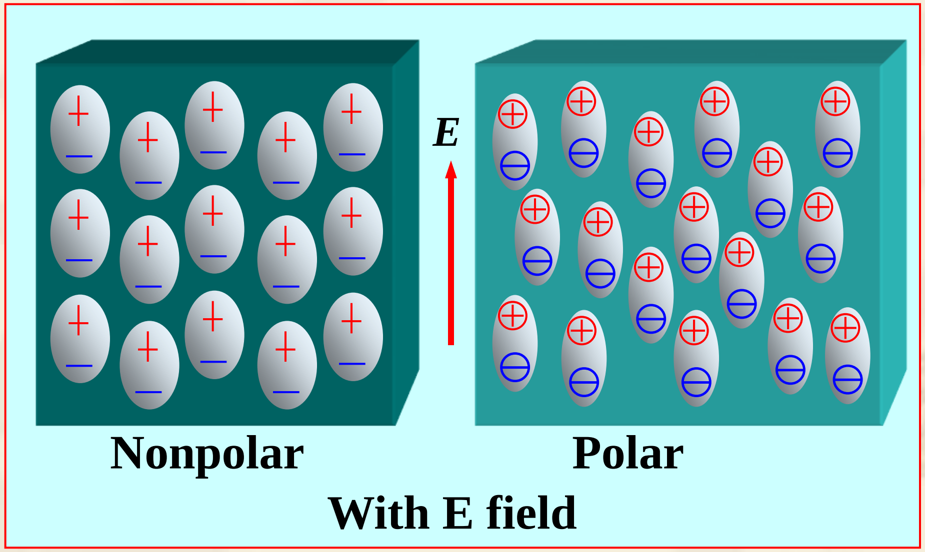




2. Polarization Intensity

Definition : **the vector sum of the electric moments per unit volume**

$$\vec{P} = \lim_{\Delta v \to 0} \frac{\sum_i \vec{p}_i}{\Delta v} \qquad (c/m^2)$$



Discussion

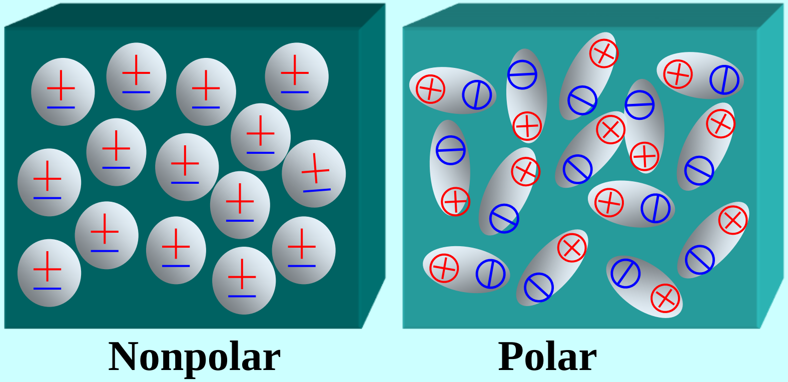
1) Without E field

- **Nonpolar molecules:**

$$\vec{p}_i = 0 \Rightarrow \vec{P} = 0$$

- **Polar molecules :**

$$\sum_i \vec{p}_i = 0 \Rightarrow \vec{P} = 0$$



No E field

2) With external E field

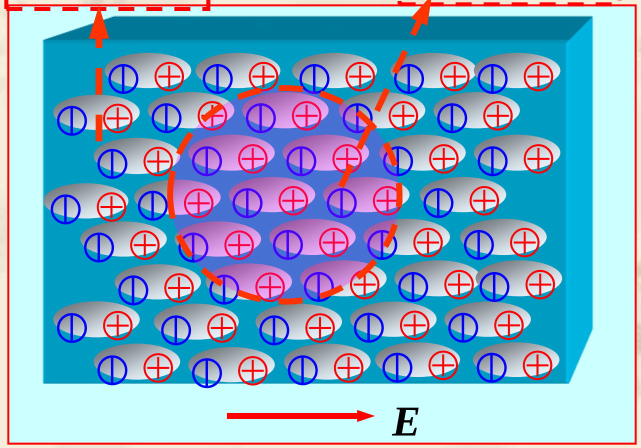
$$\vec{p}_i \neq 0 \text{ and } \vec{p}_1 = \vec{p}_2 = \cdots = \vec{p}$$

$$\Rightarrow \sum_i \vec{p}_i = N\vec{p}$$

$$\Rightarrow \vec{P} = n\vec{p}$$

n is the number of polarized molecules per unit volume

$$\vec{p} = \vec{p}_i \qquad \vec{P} = n\vec{p}$$



Conclusion

1） The magnitude of the polarization intensity is related to the dielectric material.

2） The magnitude of the polarization intensity is also related to the intensity of the applied electric field $\vec{E}_e$.

- **After the polarization, the additional electric field $\vec{E}_p$ will be generated in the space.**
- **The total electric field $\vec{E}$ in the medium is**

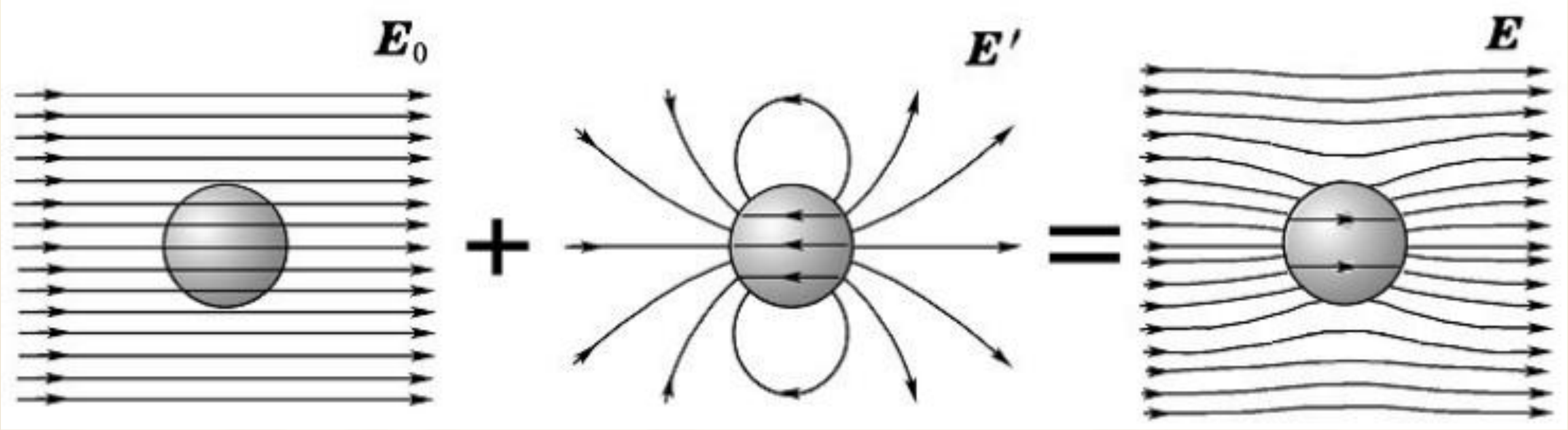
$$\vec{E} = \vec{E}_e + \vec{E}_p$$

Experimental observation shows:

For linear and isotropic media， $\vec{P}$ is proportional to $\vec{E}$

$$\vec{P} = \chi_{\mathrm{e}} \varepsilon_0 \vec{E}$$

in which, $\chi_{\mathrm{e}}(>0)$ is called the electric susceptibility



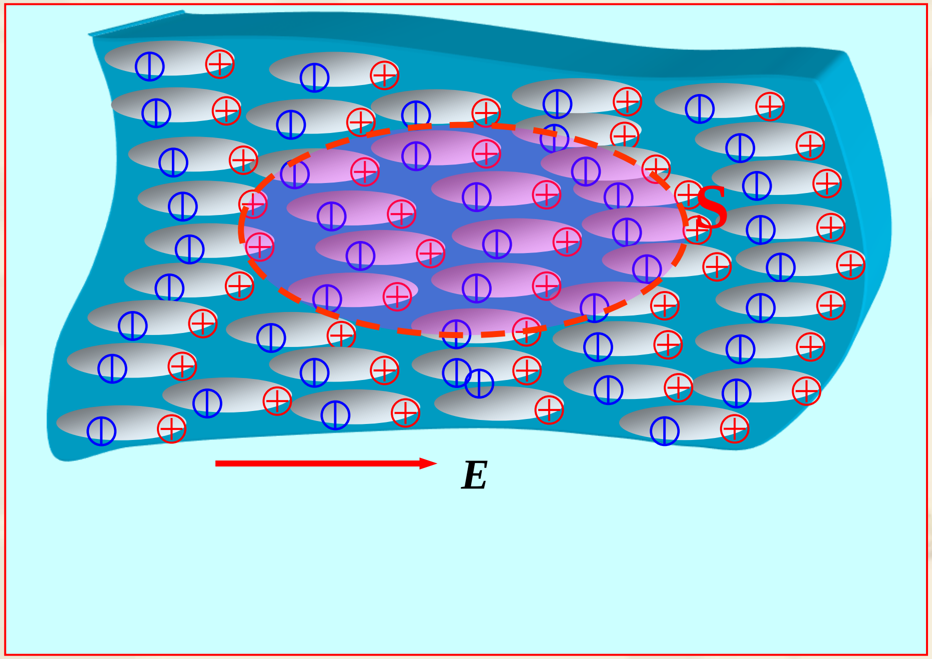
- **Additional electric field**

Inside： **the opposite direction with the applied electric field, weaker**

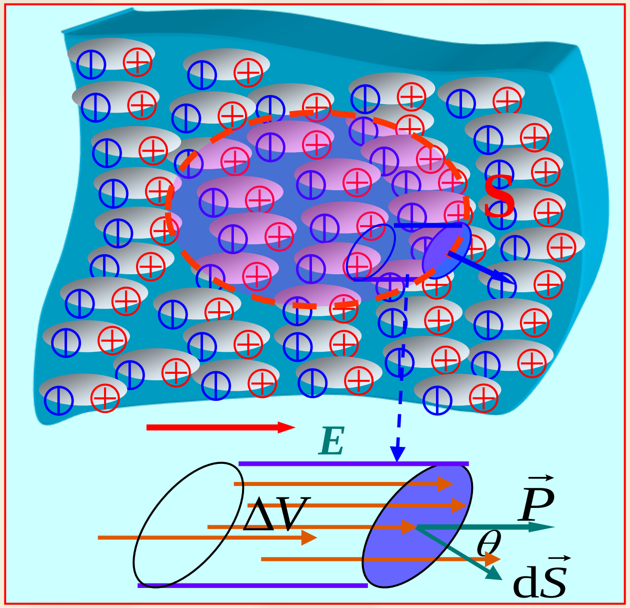
Outside： **mainly the same direction with the applied electric field, some place stronger and some place weaker**

Further discussion

- After the dielectric polarization, there might be net charges inside, which is polarized volume charge density q_P .
- After the dielectric polarization, net charges might appear on the interface, which is polarized surface charge density q_{sP} .



Calculation of polarized volume charge density



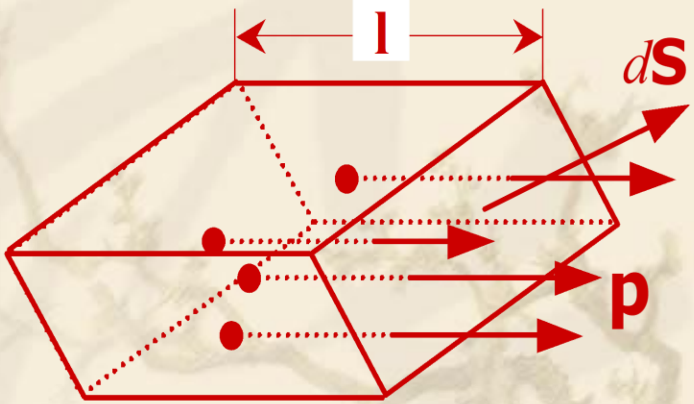
<u>Principle</u>: each polarized molecules could be considered as a dipole moment, that

$$\vec{p} = q\vec{l}$$

As shown in the figure, if the negtive charge is inside of ΔV, the corresponding positive charge is out of it.

$$q_P\big|_V = -q_P\big|_s$$

Then the total polarized charge outside of ds with the negative charges of the dipole moments in ΔV is,



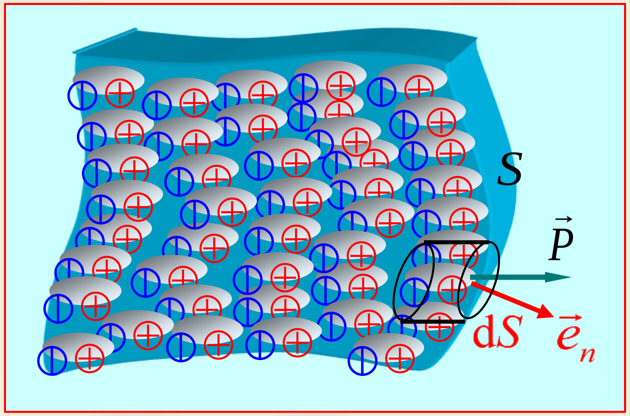
Since, $$q_P\big|_{ds} = qN_{ds} = qndV = qnl\mathrm{d}S\cos\theta$$

$$= np\mathrm{d}S\cos\theta = \vec{P}\cdot\mathrm{d}\vec{S}$$

so, $$q_P = \oint_S \vec{P}\cdot d\vec{S} = \int_V \nabla\cdot\vec{P}dV$$

$$\Rightarrow \rho_P = -\nabla\cdot\vec{P}$$

Calculation of polarized surface charge



On the interface:

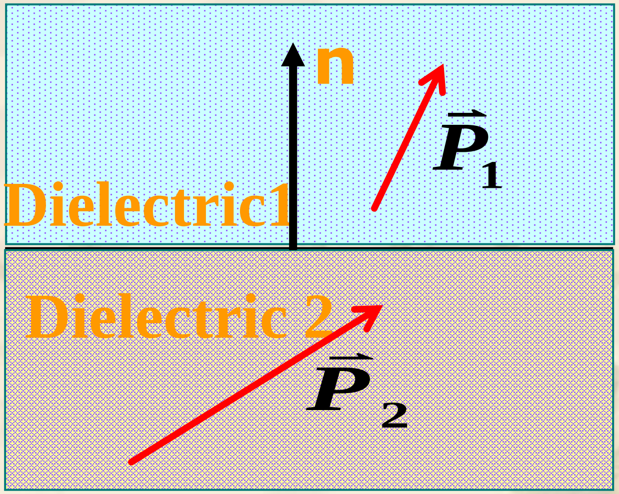
Since, $q_P\big|_{ds} = \vec{P}\cdot d\vec{S}$

so, $q_P\big|_{ds} = \vec{P}\cdot\vec{e}_n dS$

$$\Rightarrow \rho_{SP} = \vec{P}\cdot\vec{e}_n$$

Discussion: if a medium with two different types of dielectric is placed in a static filed, the polarized surface charge density should be:

$$\boxed{\rho_{SP} = \hat{n}\cdot\left(\vec{P}_2 - \vec{P}_1\right)}$$



- **The total charge of the body after polarization must remain zero**

$$\begin{aligned}\text{Total charge} &= \oint_S \rho_{ps}\, ds + \int_V \rho_p\, dv \\ &= \oint_S \mathbf{P} \cdot \mathbf{a}_n\, ds - \int_V \nabla \cdot \mathbf{P}\, dv = 0,\end{aligned}$$

3.8 Electric Flux Density and Dielectric Constant

Q: **When there is a medium in space, the polarization charge will generate an additional electric field, which affects the total electric field distribution. When calculate the total field, is it necessary to calculate the additional electric field in advance.**

$$\oint_S \vec{E}\cdot \mathrm{d}\vec{S} = \frac{1}{\varepsilon_0}(q_f + q_p)\Big|_{s内} \Rightarrow \varepsilon_0 \nabla\cdot\vec{E} = \rho_f + \rho_p$$

$$\varepsilon_0 \nabla\cdot\vec{E} = \rho_f - \nabla\bullet\vec{P}$$

$$\nabla\cdot(\varepsilon_0\vec{E} + \vec{P}) = \rho_f$$

Define Electric flux density :

$$\vec{D} = \varepsilon_0\vec{E} + \vec{P}$$

Unit: C/m^2

So

$$\nabla\cdot\vec{D} = \rho_f$$

Integral form

$$\oint_S \vec{D}\cdot \mathrm{d}\vec{S} = q_f\Big|_{s内}$$

The total outward flux of the electric flux density over any closed surface is equal to the total free charge enclosed in the surface

Conclusion

The problem of electrostatic field can be described by the following basic equations in the space of the medium :

$$\begin{cases} \nabla \cdot \vec{D} = \rho_f \\ \nabla \times \vec{E} = 0 \end{cases}$$

（Differential form ）

$$\begin{cases} \oint_S \vec{D} \cdot \mathrm{d}\vec{S} = q_f \\ \oint_C \vec{E}(\vec{r}) \cdot \mathrm{d}\vec{l} = 0 \end{cases}$$

（Integral form ）

The process of solving the problem :

$$q_f \Rightarrow \vec{D} \Rightarrow \vec{E}\ ,\ \vec{P}\ ,\ q_p$$

constitutive relationship：

$$\vec{P} = \varepsilon_0 \chi_e \vec{E} \Rightarrow \vec{D} = \varepsilon_0 (1+\chi_e)\vec{E} = \varepsilon_0 \varepsilon_r \vec{E} = \varepsilon \vec{E}$$

$$\vec{D} = \varepsilon \vec{E}$$

Dielectric constant

Relative permittivity （Dimensionless ）

❖ Classification and constitutive relationship of medium

- **Homogeneous and inhomogeneous media**
- **Isotropic and anisotropic media**
- **Time varying and time invariant medium**

- **Linear and nonlinear media**
- **Deterministic and random media**
- **Dispersive and non dispersive media**

- For anisotropic dielectrics, the relationship between the electric flux density and the electric field intensity can be written as:

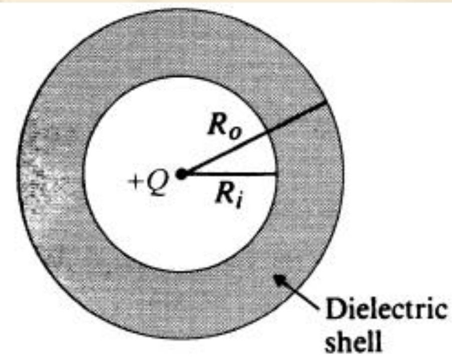
$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$

- In addition, the permittivity of a homogeneous dielectric is independent of the space coordinates.

- The relationship between the electric flux density and the electric field intensity is dependent upon the direction of applied electric field.

- The permittivity of linear dielectrics is independent of the magnitude of the electric field intensity.

EXAMPLE 3–12 A positive point charge Q is at the center of a spherical dielectric shell of an inner radius R_i and an outer radius R_o. The dielectric constant of the shell is ϵ_r. Determine **E**, V, **D**, and **P** as functions of the radial distance R.



(a)

a) $R > R_o$

$$E_{R1} = \frac{Q}{4\pi\epsilon_0 R^2} \quad D_{R1} = \epsilon_0 E_{R1} = \frac{Q}{4\pi R^2}$$

$$V_1 = \frac{Q}{4\pi\epsilon_0 R}. \qquad P_{R1} = 0.$$

b) $R_i < R < R_o$

The application of Gauss's law in this region gives us directly

$$E_{R2} = \frac{Q}{4\pi\epsilon_0\epsilon_r R^2} = \frac{Q}{4\pi\epsilon R^2}, \tag{3–110}$$

$$D_{R2} = \frac{Q}{4\pi R^2}, \tag{3–111}$$

$$P_{R2} = \left(1 - \frac{1}{\epsilon_r}\right)\frac{Q}{4\pi R^2}. \tag{3–112}$$

Note that D_{R2} has the same expression as D_{R1} and that both E_R and P_R have a discontinuity at $R = R_o$. In this region,

$$\begin{aligned} V_2 &= -\int_{\infty}^{R_o} E_{R1}\,dR - \int_{R_o}^{R} E_{R2}\,dR \\ &= V_1\Big|_{R=R_o} - \frac{Q}{4\pi\epsilon}\int_{R_o}^{R}\frac{1}{R^2}\,dR \\ &= \frac{Q}{4\pi\epsilon_0}\left[\left(1 - \frac{1}{\epsilon_r}\right)\frac{1}{R_o} + \frac{1}{\epsilon_r R}\right]. \end{aligned} \tag{3–113}$$

c) $R < R_i$

Since the medium in this region is the same as that in the region $R > R_o$, the application of Gauss's law yields the same expressions for E_R, D_R, and P_R in both regions:

$$E_{R3} = \frac{Q}{4\pi\epsilon_0 R^2},$$

$$D_{R3} = \frac{Q}{4\pi R^2},$$

$$P_{R3} = 0.$$

To find V_3, we must add to V_2 at $R = R_i$ the negative line integral of E_{R3}:

$$\begin{aligned} V_3 &= V_2\Big|_{R=R_i} - \int_{R_i}^{R} E_{R3}\,dR \\ &= \frac{Q}{4\pi\epsilon_0}\left[\left(1 - \frac{1}{\epsilon_r}\right)\frac{1}{R_o} - \left(1 - \frac{1}{\epsilon_r}\right)\frac{1}{R_i} + \frac{1}{R}\right]. \end{aligned} \tag{3-114}$$

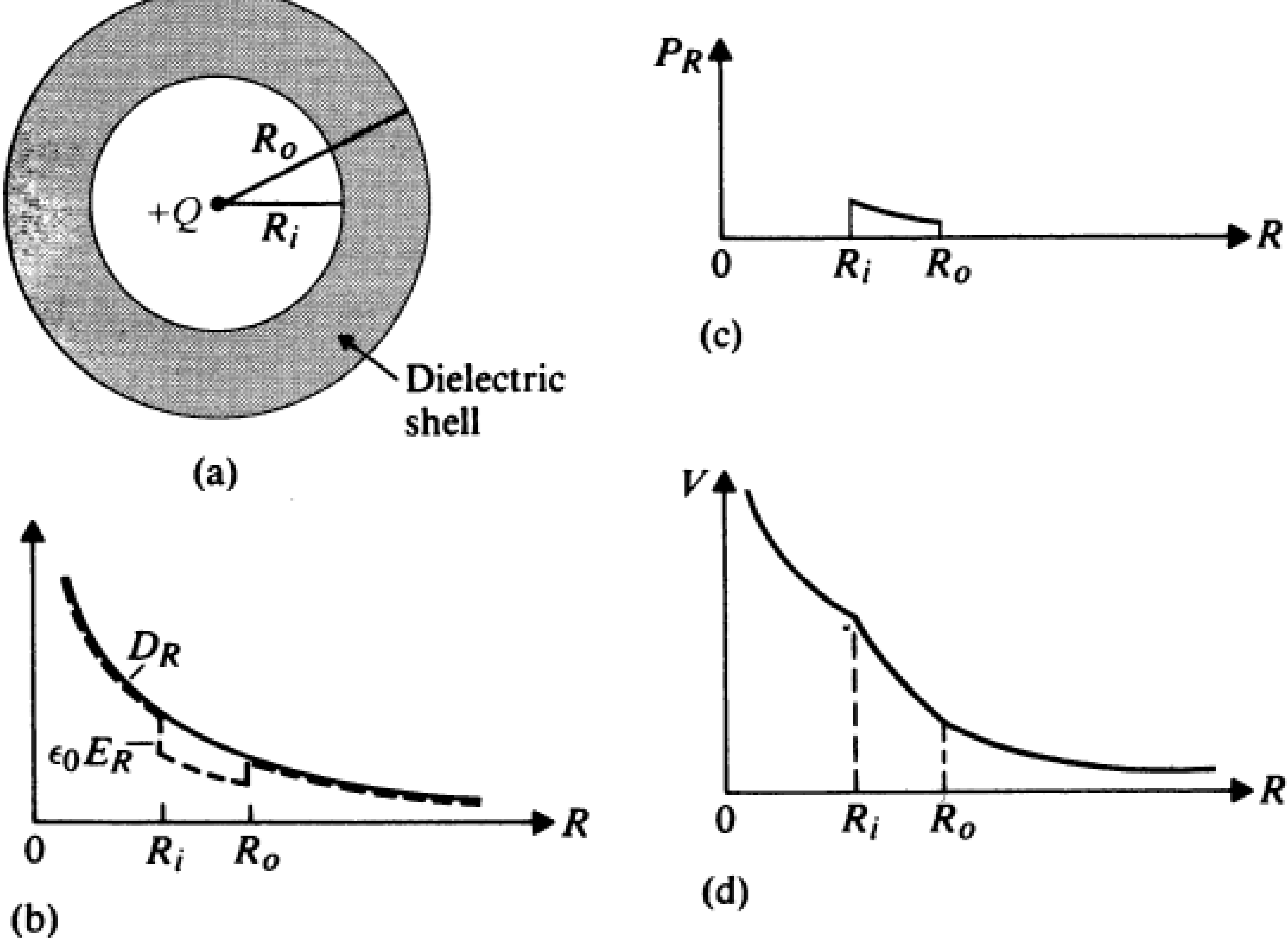


FIGURE 3–21

Field variations of a point charge $+Q$ at the center of a dielectric shell

$$\rho_{ps}\Big|_{R=R_i} = \mathbf{P}\cdot(-\mathbf{a}_R)\Big|_{R=R_i} = -P_{R2}\Big|_{R=R_i}$$
$$= -\left(1-\frac{1}{\epsilon_r}\right)\frac{Q}{4\pi R_i^2}$$

on the inner shell surface,

$$\rho_{ps}\Big|_{R=R_o} = \mathbf{P}\cdot\mathbf{a}_R\Big|_{R=R_o} = P_{R2}\Big|_{R=R_o}$$
$$= \left(1-\frac{1}{\epsilon_r}\right)\frac{Q}{4\pi R_o^2}$$

on the outer shell surface, and

$$\rho_p = -\nabla\cdot\mathbf{P}$$
$$= -\frac{1}{R^2}\frac{\partial}{\partial R}(R^2 P_{R2}) = 0.$$

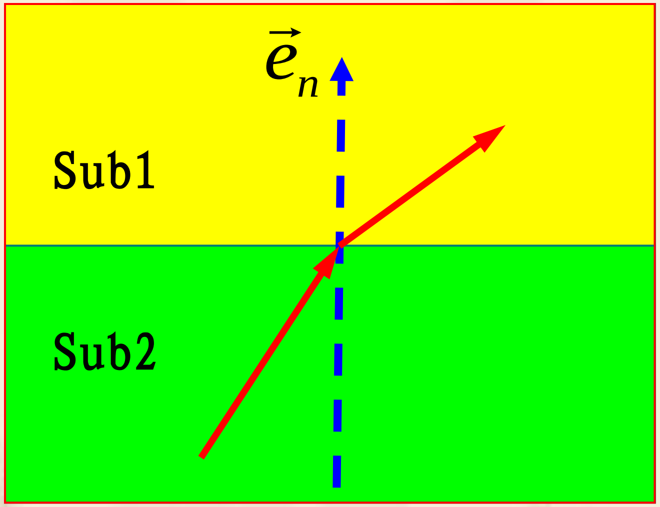
- **There is no net polarization volume charge inside the dielectric shell.**
- **Negative polarization surface charges exist on the inner surface and positive polarization surface on the outer surface.**

3.9 Boundary Conditions for Electrostatic Fields

- **What is the boundary condition?**

- **Why to study BC?**

- **How to discuss BC?**



A real EM question is come forward in terms of a specific space. The space may be composed by different media. BC outline the relationship that are satisfied by the EM vectors on that boundary surface between any two different media, that is the basic quality for the EM field on the surface of different media.

Physical: due to the abrupt change of the characteristic parameters on the both sides of an interface, the field on the two side will change as well. The integral form of the Maxwell's equation lost meaning on the both sides of the interface, the BC is necessary.

Mathematics: Maxwell's equations are integral functions, the solution of them are not determinative, the BC play the role of identify the determinative solution

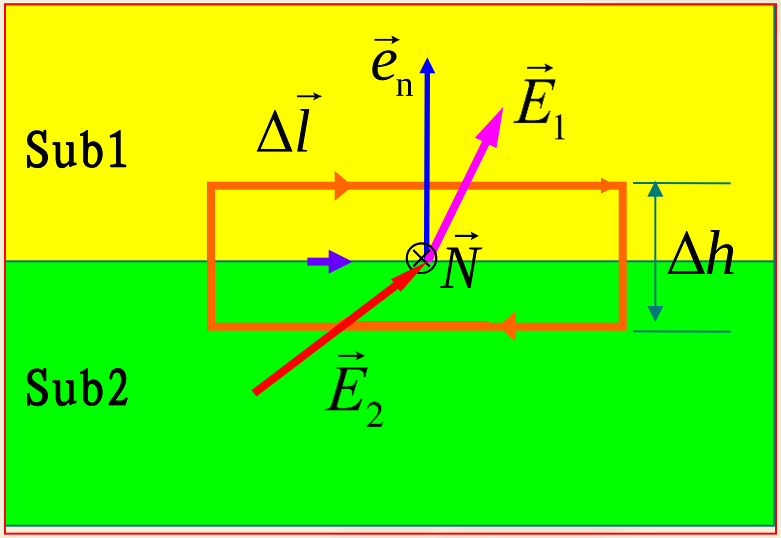
The **integral form** of Maxwell's equation is still suitable for the surface of different media, which is helpful derive the expression of the BC in terms of different media

(1) The tangential directional BC

On the dual sides of the interface of medias, selecting a small loop as the graph herein, and setting $\Delta h \to 0$, Since

$$\oint_c \vec{E} \cdot dl = \vec{E}_1 \cdot \Delta l + \vec{E}_2 \cdot (-\Delta l) = E_{1t}\Delta l - E_{2t}\Delta l = 0$$

$$\Rightarrow \quad \vec{e}_{\mathrm{n}} \times (\vec{E}_1 - \vec{E}_2) = 0 \qquad \textbf{Or} \qquad E_{1\mathrm{t}} = E_{2\mathrm{t}}$$

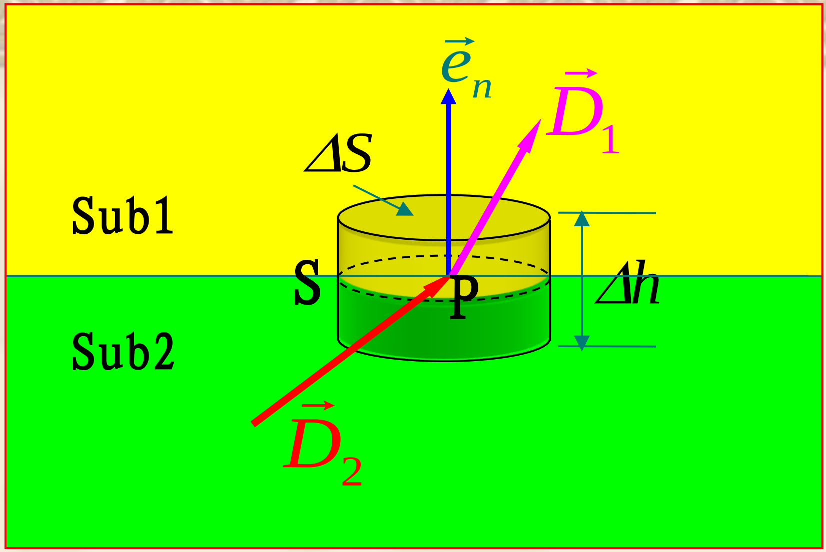


- **The tangential component of an E field is continuous across an interface**

$$\frac{D_{1t}}{\varepsilon_1} = \frac{D_{2t}}{\varepsilon_2}$$

(2) The normal directional BC

Defining a point *P* on the interface between two medias, and drawing a flat cylinder surface *S* enclosing the point.



$\Delta h \to 0$, then

$$\oint_S \vec{D} \cdot \mathrm{d}\vec{S} = q \quad \Rightarrow \quad (\vec{D}_1 - \vec{D}_2) \cdot \vec{e}_\mathrm{n} \Delta S = \rho_S \Delta S$$

so $$\vec{e}_\mathrm{n} \cdot (\vec{D}_1 - \vec{D}_2) = \rho_S$$

or $$\boldsymbol{D}_{1\mathrm{n}} - \boldsymbol{D}_{2\mathrm{n}} = \rho_S$$

if $$\rho_S = 0$$

$$\Rightarrow \quad D_{1n} = D_{2n}$$

$$\Rightarrow \quad \varepsilon_1 E_{1n} = \varepsilon_2 E_{2n}$$

- **The normal component of D field is discontinuous across an interface where a surface charge exists;**
- **the amount of discontinuity being equal to the surface charge density.**

❖ Basic equation (general problems)

Differential forms :
$$\begin{cases}\nabla\cdot\vec{\mathbf{D}}=\rho\\ \nabla\times\vec{\mathbf{E}}=0\end{cases}$$

Integral forms :
$$\begin{cases}\oint_S\vec{\mathbf{D}}\cdot d\vec{\mathbf{S}}=q\\ \oint_C\vec{\mathbf{E}}\cdot d\vec{\mathbf{l}}=0\end{cases}$$

Constitutive relations:
$$\vec{D}=\varepsilon\vec{E}$$

❖ Boundary conditions (general problems)

$$\begin{cases}\vec{e}_{\mathrm{n}}\cdot(\vec{\mathbf{D}}_1-\vec{\mathbf{D}}_2)=\rho_S\\ \vec{e}_{\mathrm{n}}\times(\vec{\mathbf{E}}_1-\vec{\mathbf{E}}_2)=0\end{cases}$$

Or

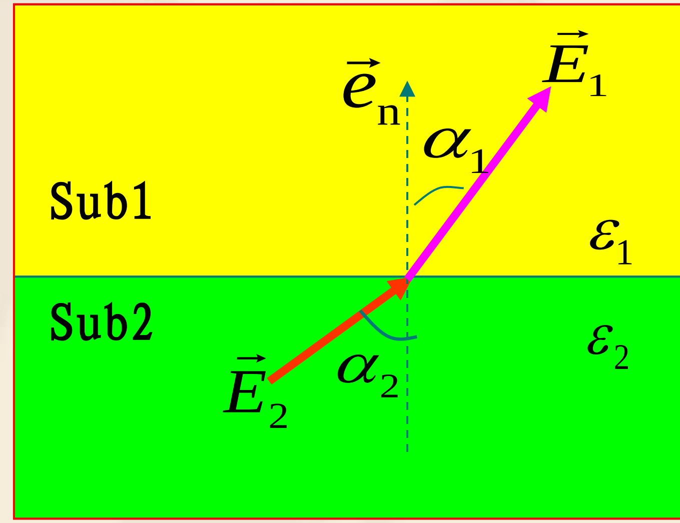
$$\begin{cases}D_{1\mathrm{n}}-D_{2\mathrm{n}}=\rho_S\\ E_{1\mathrm{t}}-E_{2\mathrm{t}}=0\end{cases}$$

❖ Special problems According to the classification of medium

- **Perfect dielectric** $\sigma=0$
- **conductor** $\sigma\neq0$

Ideal dielectric

- The natural boundary conditions on the dielectric surface
$$\begin{cases} D_{1n} - D_{2n} = 0 \\ E_{1t} - E_{2t} = 0 \end{cases}$$



- Direction relation of the field vector at the two sides of the interface

$$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{E_{1t}/E_{1n}}{E_{2t}/E_{2n}} = \frac{\varepsilon_1/D_{1n}}{\varepsilon_2/D_{2n}} = \frac{\varepsilon_1}{\varepsilon_2}$$

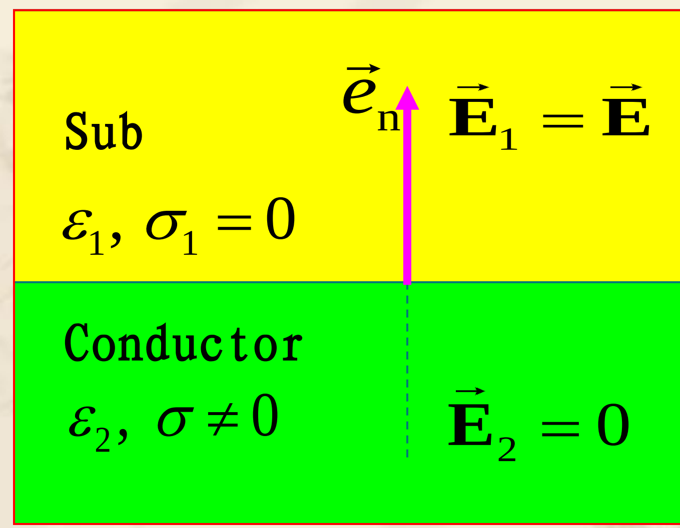
Conductor

- Electrostatic balance

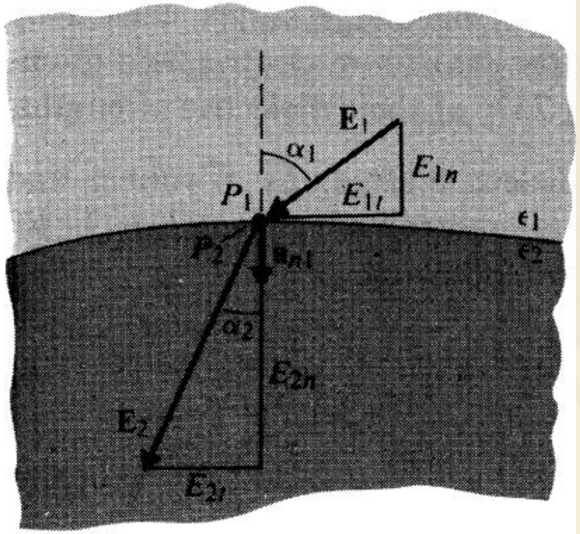
The electric field inside the conductor is **zero**

- Boundary conditions on the surface of a conductor

$$\begin{cases} \vec{e}_n \cdot \vec{\mathbf{D}} = \rho_S \\ \vec{e}_n \times \vec{\mathbf{E}} = 0 \end{cases} \quad \text{OR} \quad \begin{cases} D_n = \rho_S \\ E_t = 0 \end{cases}$$



EXAMPLE 3–15 Two dielectric media with permittivities ϵ_1 and ϵ_2 are separated by a charge-free boundary as shown in Fig. 3–25. The electric field intensity in medium 1 at the point P_1 has a magnitude E_1 and makes an angle α_1 with the normal. Determine the magnitude and direction of the electric field intensity at point P_2 in medium 2.



Solution: **there is no free charge on the surface, from the boundary condition**

$$\begin{cases} E_{1\mathrm{t}} = E_{2\mathrm{t}} \\ \boldsymbol{D}_{1\mathrm{n}} - \boldsymbol{D}_{2\mathrm{n}} = 0 \end{cases} \Rightarrow \begin{cases} E_1 \sin\alpha_1 = E_2 \sin\alpha_2 \\ \varepsilon_1 E_1 \cos\alpha_1 = \varepsilon_2 E_2 \cos\alpha_2 \end{cases}$$

$$\Rightarrow \frac{\tan\alpha_2}{\tan\alpha_1} = \frac{\varepsilon_2}{\varepsilon_1}$$

Q: Whether ε_1 is large or smaller than ε_2?

$$E_2 = \sqrt{E_{2t}^2 + E_{2n}^2} = \sqrt{(E_2 \sin\alpha_2)^2 + (E_2 \cos\alpha_2)^2}$$

$$\Rightarrow = [(E_1 \sin\alpha_1)^2 + (\frac{\varepsilon_1}{\varepsilon_2} E_1 \cos\alpha_1)^2]^{1/2}$$

$$= E_1[(\sin\alpha_1)^2 + (\frac{\varepsilon_1}{\varepsilon_2} \cos\alpha_1)^2]^{1/2}$$

- **Boundary conditions of electrostatic potential (arbitrary electrostatic field)**

$$\begin{cases} D_{1n} - D_{2n} = \rho_S \\ E_{1t} - E_{2t} = 0 \end{cases} \xrightarrow{\vec{\mathbf{D}} = -\varepsilon \nabla \phi} \varepsilon_2 \frac{\partial \phi_2}{\partial n} - \varepsilon_1 \frac{\partial \phi_1}{\partial n} = \rho_S$$

$$\frac{\partial}{\partial v}(\varphi_1 - \varphi_2) = 0 \qquad \varphi_1 - \varphi_2 = c \quad \Rightarrow \quad \phi_1 = \phi_2$$

- **Typical conditions for electrostatic field**

  - **The interface between two perfect medium (No free charge)**

$$\rho_S = 0 \Rightarrow \varepsilon_2 \frac{\partial \phi_2}{\partial n} = \varepsilon_1 \frac{\partial \phi_1}{\partial n}$$

- **On the surface of conductors**

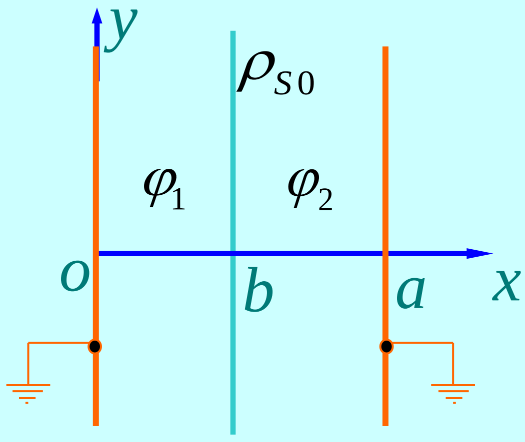
$\phi =$ **constant ,** $\varepsilon \frac{\partial \phi}{\partial n} = -\rho_S$

Ex **Two infinite grounded conductor planes are located at x=0 and x=a, there is a uniformly charged plane at x=b with surface charge density of ρ_{S0} . Find the electric potential and electric field intensity between these two plane.**

Solution: $\varphi_1 = \varphi_1(x)\,, \qquad \varphi_2 = \varphi_2(x)$

$$\frac{\mathrm{d}^2\varphi_1(x)}{\mathrm{d}x^2} = 0, \qquad (0 < x < b)$$

$$\frac{\mathrm{d}^2\varphi_2(x)}{\mathrm{d}x^2} = 0, \qquad (b < x < a)$$



Two infinite plane

So $\varphi_1(x) = C_1 x + D_1$

$$\varphi_2(x) = C_2 x + D_2$$

Determine the coefficient **Applying the B.C.**

$x = 0 \quad \varphi_1(0) = 0$

$\varphi_2(a) = 0$

$x=b \quad \varphi_1(b) = \varphi_2(b),$

$$\left[\frac{\partial \varphi_2(x)}{\partial x} - \frac{\partial \varphi_1(x)}{\partial x}\right]_{x=b} = -\frac{\rho_{S0}}{\varepsilon_0}$$

$$\Rightarrow$$

$$C_1 = -\frac{\rho_{S0}(b-a)}{\varepsilon_0 a}, \quad D_1 = 0$$

$$C_2 = -\frac{\rho_{S0} b}{\varepsilon_0 a}, \quad D_2 = \frac{\rho_{S0} b}{\varepsilon_0}$$

So

$$\varphi_1(x) = \frac{\rho_{S0}(a-b)}{\varepsilon_0 a} x, \quad (0 \leqslant x \leqslant b)$$

$$\vec{E}_1(x) = -\nabla \varphi_1(x) = -\vec{e}_x \frac{\rho_{S0}(a-b)}{\varepsilon_0 a}$$

$$\varphi_2(x) = \frac{\rho_{S0} b}{\varepsilon_0 a}(a-x), \quad (b \leqslant x \leqslant a)$$

$$\vec{E}_2(x) = -\nabla \varphi_2(x) = \vec{e}_x \frac{\rho_{S0} b}{\varepsilon_0 a}$$

Question:

Shall it be solved by electric intensity?

Method Two

The three boards might be considered as two parallel plates capacitors, and the electric intensity should direct to $\vec{e}_x$

So we assume $\vec{E}_1 = \vec{e}_x E_1, \vec{E}_2 = \vec{e}_x E_2$

Find boundary conditions:

$$E_2 - E_1 = \frac{\rho_{S0}}{\varepsilon_0}$$

Gauss's Law

$$-E_1 b = E_2 (a - b)$$

According to B.C. of petential

$$\Rightarrow \begin{cases} \vec{E}_1(x) = -\vec{e}_x \dfrac{\rho_{S0}(a-b)}{\varepsilon_0 a} & 0 \le x \le b \\ \vec{E}_2(x) = \vec{e}_x \dfrac{\rho_{S0} b}{\varepsilon_0 a} & b \le x \le a \end{cases}$$

$$\Rightarrow \begin{cases} \varphi_1(x) = \int_x^0 \vec{E}_1 \bullet d\vec{x} = \dfrac{\rho_{S0}(a-b)}{\varepsilon_0 a} x & 0 \le x \le b \\ \varphi_2(x) = \int_x^a \vec{E}_2 \bullet d\vec{x} = \dfrac{\rho_{S0} b}{\varepsilon_0 a}(a - x) & b \le x \le a \end{cases}$$

3.10 Capacitance and Capacitors

- A conductor in an electrostatic field is an equipotential body.
- Suppose the potential due to a charge Q is U. Increasing the total charge will increase the potential as well;
- The ratio Q/U remains unchanged.

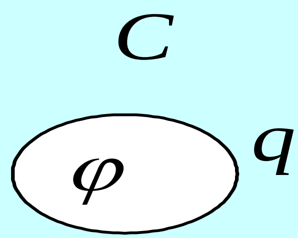
$$Q = CU$$

- Capacitors can have advantage such as Storage, filtering, phase shift, blocking, bypass, frequency selection etc.
- Capacitors can have disadvantage such as electromagnetic compatibility problems

Capacitance

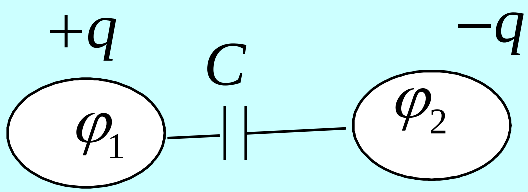
- **Isolated conductor capacitance**

$$C = \frac{q}{\phi}$$



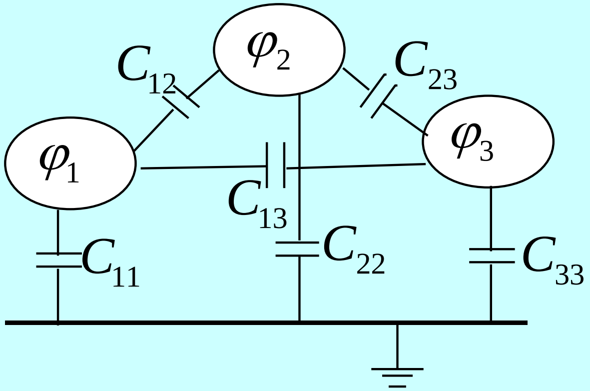
- **Capacitance of the capacitor composed of two conductors**

$$C = \frac{q}{U} = \frac{q}{|\varphi_1 - \varphi_2|}$$



- **Conductors form part of the capacitor system in a multi-conductor system**

$$\begin{cases} q_1 = C_{11}\varphi_1 + C_{12}(\varphi_1 - \varphi_2) + C_{13}(\varphi_1 - \varphi_3) \\ q_2 = C_{22}\varphi_2 + C_{21}(\varphi_2 - \varphi_1) + C_{23}(\varphi_2 - \varphi_3) \\ q_3 = C_{33}\varphi_3 + C_{31}(\varphi_3 - \varphi_1) + C_{32}(\varphi_3 - \varphi_2) \end{cases}$$



The factors to determine The capacitance

- Depends on the **structure**, **size**, **shape** and the surrounding **dielectric** of the conductor system.

- Independent to the **charge** and the **electric potential** of the conductor.

Method for solving capacitance

（Electrical capacitance is independent of the charge quantity and potential of the conductor）

Method One :

✓Assume charge q / $\pm q$ of the conductor/ between two conductors

✓Find the potential / voltage of the conductor/ between two conductors

Method two:

✓Assume the potential / voltage of the conductor/ between two conductors

✓Solve the charge q on the conductor surface .

According to the definition for capacitance

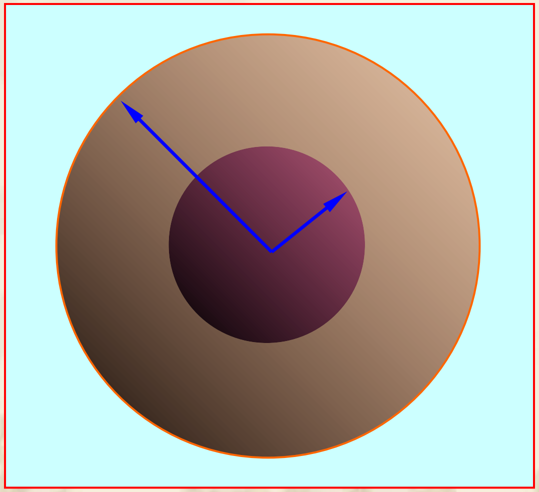
$$C = Q/U$$

Ex: Concentric spherical capacitor with inner conductor radius of a and outer conductor radius of b, homogeneous medium with ε Find the Capacitance of the spherical capacitor.

Solution： Charge q ， according to Gauss theorem

$$\vec{D} = \vec{e}_r \frac{q}{4\pi r^2}, \qquad \vec{E} = \vec{e}_r \frac{q}{4\pi \varepsilon r^2}$$

The voltage between the concentric conductor:



$$U = \int_a^b E\mathrm{d}r = \frac{q}{4\pi\varepsilon}\left(\frac{1}{a} - \frac{1}{b}\right) = \frac{q}{4\pi\varepsilon} \cdot \frac{b-a}{ab}$$

Capacitance:

$$C = \frac{q}{U} = \frac{4\pi\varepsilon ab}{b-a}$$

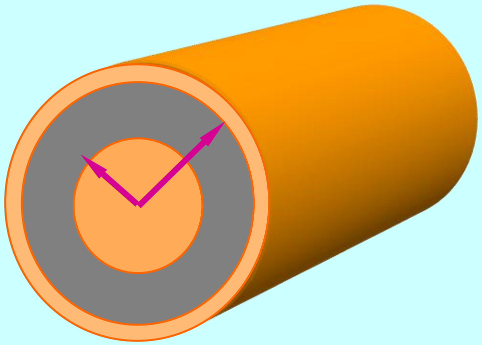
$$b \to \infty \qquad C = 4\pi\varepsilon a$$

Capacitance for Isolated conductor ball

Ex: **Coaxial line (of infinite length) with inner conductor radius a and outer conductor radius b, a dielectric of permittivity εis filled up in between. Find the capacitance per unit length.**

Solution: **assume the line charge density is $+\rho_l$ and $-\rho_l$ for the inner and outer conductor, applying Gauss's Law and the electric field between two conductor is:**

$$\vec{E}(\rho) = \vec{e}_\rho \frac{\rho_l}{2\pi\varepsilon\rho}$$



Potential difference between inner and outer conductor

$$U = \int_a^b \vec{E}(\rho)\cdot\vec{e}_\rho \mathrm{d}\rho = \frac{\rho_l}{2\pi\varepsilon}\int_a^b \frac{1}{\rho}\mathrm{d}\rho$$
$$= \frac{\rho_l}{2\pi\varepsilon}\ln(b/a)$$

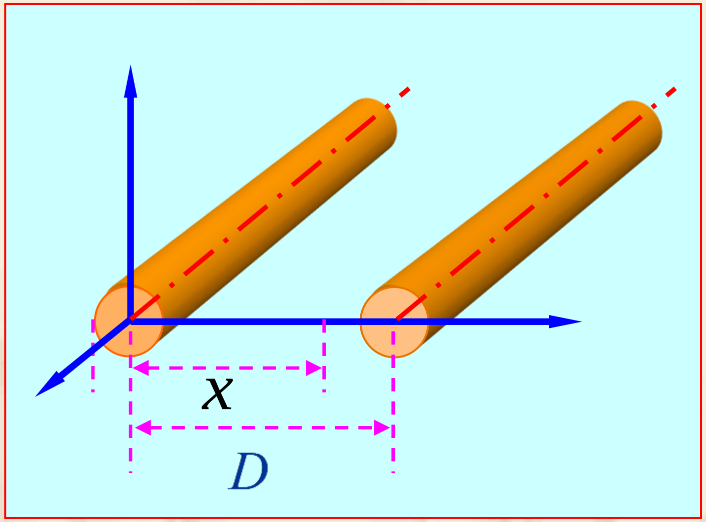
the capacitance per unit length:

$$C_1 = \frac{\rho_l}{U} = \frac{2\pi\varepsilon}{\ln(b/a)} \qquad \text{(F/m)}$$

Ex: The parallel dual-transmission line with the conductor radius of a, and distance of D ($D >> a$), find the capacitance per unit length of this system

Solution: assume the line charge density is $+\rho_l$ and $-\rho_l$ for two conductors, since $D >> a$, the charge is considered to be uniformly distributed on the surface, and the electric field between two conductor is:

$$\vec{E}(x) = \vec{e}_x \frac{\rho_l}{2\pi\varepsilon_0}\left(\frac{1}{x} + \frac{1}{D-x}\right)$$



Potential difference between two conductors is:

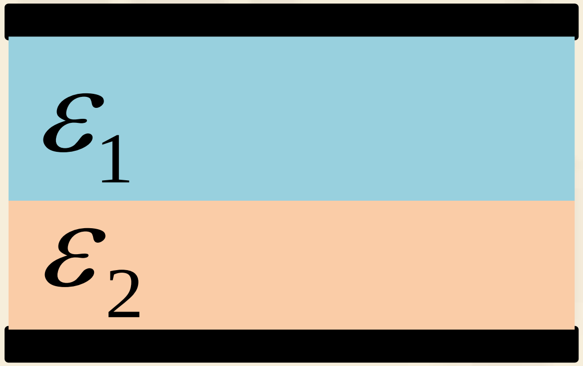
$$U = \int_1^2 \vec{E} \cdot \mathrm{d}\vec{l}$$

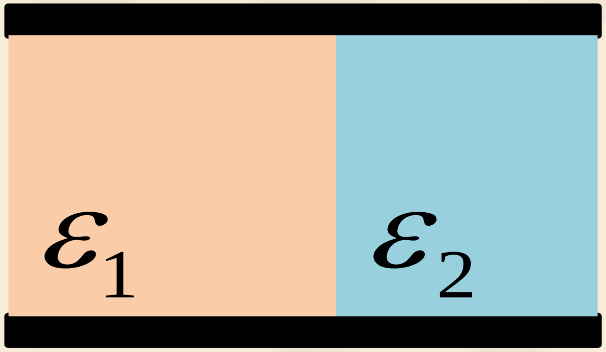
$$= \frac{\rho_l}{2\pi\varepsilon_0} \int_a^{D-a} \left(\frac{1}{x} + \frac{1}{D-x}\right) \mathrm{d}x = \frac{\rho_l}{\pi\varepsilon_0} \ln \frac{D-a}{a}$$

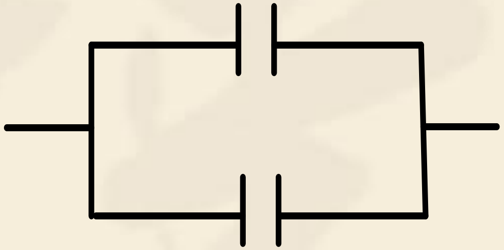
the capacitance per unit length:

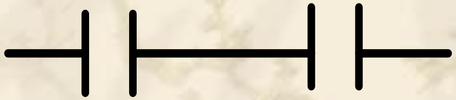
$$C_1 = \frac{\rho_l}{U} = \frac{\pi\varepsilon_0}{\ln[(D-a)/a]} \approx \frac{\pi\varepsilon_0}{\ln(D/a)} \quad \text{(F/m)}$$

Discussion：draw the equal circuit (connection of capacitors) of the following structures









3.11 Electrostatic Energy

- Under the influence of electrostatic fields, a positively charged body will be moved along the direction of the electric field. In this case, work is being done by the field.
- In order to do this work the electrostatic field must lose energy, implying that the electrostatic field has stored energy.
- If the charged body is moved into the electrostatic field from infinity by an applied force, work is being done against the electric force.
- This work will become part of the energy stored in the field, and the total energy of the electrostatic field will be increased.

$$W_e = \int_0^Q \varphi(q)\,\mathrm{d}q$$

- The total energy stored in an electrostatic field can be determined from the work done to assemble the charges giving rise to the field.

- **Calculate the energy of an isolated body with charge *Q*. Assuming that the charge *Q* *with element* d*q* was moved in from infinity.**

- Since there was no field in space at the beginning, the applied force did not need to do the work to move in the first element d*q*.

- When the element dq that followed was moved in, the applied force had to do the work against the electric field.

- Let the final electric potential of the body be Φ and the final charge on it be Q. At the moment when the charge of the body is increased to $q=\alpha Q,\ 0<\alpha<1$, the electric potential of the charged body is then $\varphi=\alpha\Phi$.

- In this way, when the charges of all bodies are increased by the same ratio α, the work by the applied force, and hence the incremental electrostatic energy of the charged system is

$$\mathrm{d}W_{\mathrm{e}}=\varphi\,\mathrm{d}q=\Phi Q\alpha\,\mathrm{d}\alpha$$

So the toal electrostatic energy should be calculated as,

$$W_\mathrm{e} = \int \mathrm{d}W_\mathrm{e} = \Phi Q \int_0^1 \alpha \mathrm{d}\alpha = \frac{1}{2}\Phi Q$$

Applying $\varphi = \dfrac{q}{C}$ **We also get** $W_\mathrm{e} = \dfrac{1}{2}\dfrac{Q^2}{C}$

If there are n bodiescharged with $Q_1, Q_2, \cdots, Q_n$ **, the total energy of the system is given by**

$$W_\mathrm{e} = \sum_{i=1}^{n} \frac{1}{2} \Phi_i Q_i$$

If the charge is distributed in a volume, on a surface, or at a line, considering $\mathrm{d}q = \rho\,\mathrm{d}V = \rho_S\mathrm{d}S = \rho_l\mathrm{d}l$, then the total energy of the body with the distributed charge can be written accordingly as

$$W_\mathrm{e} = \int_V \frac{1}{2}\varphi\,\rho\,\mathrm{d}V = \int_S \frac{1}{2}\varphi\,\rho_S\,\mathrm{d}S = \int_l \frac{1}{2}\varphi\,\rho_l\,\mathrm{d}l$$

Where φ is the electric potential at the element volume d*V*, the element surface d*S*, or the element line d*l* , respectively.

From the point of view of the field, the electrostatic energy is distributed in the entire space where the electric field is found. We now discuss how to calculate the energy density of electro-static field.

The energy density of electrostatic field is denoted by small letter character w_e.

Suppose the charges distributing in volume V is *Q*.

Integral will be unchanged if the integral is in a infinite sphere

$$W_{\mathrm{e}} = \frac{1}{2}\int_V \phi\rho dv = \frac{1}{2}\int_V \phi(\nabla\cdot\vec{D})dv$$

$$= \frac{1}{2}\int_V \nabla\cdot(\phi\vec{D})dv - \frac{1}{2}\int_V \nabla\phi\cdot\vec{D}dv$$

$$= \frac{1}{2}\oint_S \phi\vec{D}\cdot d\vec{s} + \frac{1}{2}\int_V \vec{E}\cdot\vec{D}dv$$

$$= \frac{1}{2}\int_V \vec{E}\cdot\vec{D}dv$$

$$\oint_{S_\infty} \varphi\,\boldsymbol{D}\cdot\mathrm{d}\boldsymbol{S} \to 0$$

$$\oint_S \varphi\vec{D}\cdot\mathrm{d}\vec{S} \sim \mathrm{O}\left(\int_S \frac{1}{R}\cdot\frac{1}{R^2}\mathrm{d}S\right) \sim \mathrm{O}\left(\frac{1}{R}\right) \to 0$$

We can see that the energy density of the electrostatic field is

$$w_{\mathrm{e}} = \frac{1}{2}\boldsymbol{D}\cdot\boldsymbol{E}$$

For linear isotropic dielectrics, $D = \varepsilon E$, we have

$$w_e = \frac{1}{2}\varepsilon E^2$$

- The energy of an electrostatic field is proportional to the square of the magnitude of the field intensity. Consequently, the energy does not obey the principle of superposition.

- The total energy is not equal to the sum of their energies when they exist on their own.

Reason: When the second charged body is introduced, work needs to be done against the electric field due to the first charged body, and this work done becomes the energy stored in the field. This energy is called mutual energy, while the energy is called self energy when the charged body is alone.

- ## Calculated by potential

✓**Distribution of Body Charge** $W_e = \frac{1}{2}\int_V \rho\phi \mathrm{d}V$

✓**Distribution of plane Charge** $W_e = \frac{1}{2}\int_S \rho_S \phi \mathrm{d}S$

✓**Storage of Capacitor energy**

$$W_e = \frac{1}{2}\sum_i \oint_{S_i} \rho_{S_i}\phi_i \mathrm{d}S = \frac{1}{2}\sum_i \phi_i \oint_{S_i} \rho_{S_i} \mathrm{d}S = \frac{1}{2}\sum_i \phi_i q_i$$

Where: q_i —— **Charge of the i conductors**

φ_i —— **potential of the i conductors**

● Calculated by distribution

- ✓ energy density of electric field： $w_{\mathrm{e}} = \dfrac{1}{2}\vec{D}\cdot\vec{E}$

- ✓ Total energy of electric field： $W_{\mathrm{e}} = \dfrac{1}{2}\int_V \vec{D}\cdot\vec{E}\,\mathrm{d}V$

Integral region is the whole space of the electric field

- ✓ For linear and isotropic medium

$$w_{\mathrm{e}} = \frac{1}{2}\vec{D}\cdot\vec{E} = \frac{1}{2}\varepsilon\vec{E}\cdot\vec{E} = \frac{1}{2}\varepsilon E^2$$

$$W_{\mathrm{e}} = \frac{1}{2}\int_V \vec{D}\cdot\vec{E}\,\mathrm{d}V = \frac{1}{2}\int_V \varepsilon\vec{E}\cdot\vec{E}\,\mathrm{d}V = \frac{1}{2}\int_V \varepsilon E^2\,\mathrm{d}V$$

Ex. Find the energy of a conducting sphere with radius a and charge Q. The permittivity of the dielectric around the conductor is ε.

Solution: Three methods can be used.

(1) The electric potential of a conducting sphere with radius a and charge Q is

$$\varphi = \frac{Q}{4\pi\varepsilon a}$$

We find

$$W_{\mathrm{e}} = \frac{1}{2}\Phi Q = \frac{Q^2}{8\pi\varepsilon a}$$

(2) Since the surface of a conductor is an equipotential surface, and

$$W_{\mathrm{e}} = \frac{1}{2}\oint_S \frac{Q}{4\pi\varepsilon a}\rho_S \mathrm{d}S = \frac{Q^2}{8\pi\varepsilon a}$$

(3) The electric field intensity caused by a conducting sphere with charge Q is

$$E = \frac{Q}{4\pi\varepsilon r^2}$$

And the energy density is

$$w_e = \frac{Q^2}{32\pi^2\varepsilon r^4}$$

Integrating over the region outside the conductor gives

$$W_e = \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \int_a^{\infty} w_e r^2 \sin\theta \, dr = \frac{Q^2}{8\pi\varepsilon a}$$

Three results above are in agreement !

Homework

3-5, 3-11, 3-12, 3-22, 3-25, 3-33, 3-37, 3-40